Declining Fertility and Housing Costs

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Abstract

I study how the cost of family space shapes fertility decisions, and what role housing policy can play. Housing costs are at the center of the debate on declining fertility: children need space, and the young households who are deciding whether to have kids have few resources to pay for it. I build a life-cycle model in which households face income risk and choose fertility, savings, and whether to rent or own, subject to a down payment constraint. In the model, a child increases the demand for physical space, and young households face a tradeoff between having children sooner, when fecundity is high but earnings are low and uncertain, and later, after building up savings. I calibrate the model to the United States in 2007 and estimate an exogenous decrease in the desire for children to match the recent fertility decline. Along the transition to a new steady state with a smaller population, the presence of fewer households makes housing cheaper, which feeds back into higher fertility. I use the estimated model to assess different housing policies. Relaxing credit requirements has small effects on fertility in the medium run, as it mostly shifts the timing of fertility. Instead, the effect of policies that change long-term incentives, such as more housing supply and credit available only to parents, is stronger, and builds up over time. The results shed light on the importance of focusing on the long-run effects of policies aimed at increasing fertility.

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1 Introduction

Fertility is falling almost everywhere. In the United States, the period fertility rate fell from about 2.1 children per woman in 2007 to 1.62 in 2023, and by some estimates the world as a whole is now below replacement for the first time (World Bank, 2026; Fernández-Villaverde and Norrick, 2026). The consequences of this decline are vast, ranging from pressure on social security to slower growth. As a result, this phenomenon has become a central concern for policymakers. The causes of the decline are elusive,[1] but housing is often at the center of the discussion on how to raise fertility in the future, both in public debate and in policy.[2] The logic is intuitive. Housing is the largest item in household budgets, representing a third of all spending of the average US household (U.S. Bureau of Labor Statistics, 2025). Children increase the pressure on this already high bill: they require physical space, so becoming a parent raises a household's demand for housing. On the other hand, prospective parents are young, with low earnings and savings, so that the extra room is particularly expensive exactly when it could shift the fertility decision.

Many policy proposals seek to help young families buy a home, intervening on the financial constraints by lowering the cost of borrowing or the down payment. If saving for a down payment is what keeps young couples out of a 'family-sized' home, especially where larger homes have to be bought,[3] such policies could raise births. However, the extent to which this is quantitatively relevant is unclear. Easier credit lowers the down payment, but not the cost of the extra space, which a family pays every year its children live at home: whether it rents or owns, and in bad years as well as good ones. Looser credit may therefore lead young couples to buy a home, and perhaps to have children earlier, without lowering the long-run cost of children enough to make them have more.

This paper asks how the cost of family space shapes fertility, and what housing policy can do about it. I answer this question with a quantitative life-cycle model in which households face idiosyncratic earnings risk and choose whether to have children, whether to rent or own, how much housing to demand and how much to save. I find that housing policy raises births most when it lowers the long-run cost of having children, through cheaper space or credit available only to parents, rather

[1] The literature has not reached a consensus. For example, Kearney et al. (2022) do not find a single economic or policy factor that accounts for the decline. Kearney and Levine (2026) argue for "shifting priorities" across cohorts, a change in norms around work, parenting, gender roles and leisure, shaped in part by new technologies such as the smartphone and social media. Doepke et al. (2023) argue that in rich countries fertility now depends on how easily parents can combine careers and family.

[2] In a fourteen-country survey, housing is among the most cited reasons for having fewer children than desired, after finances and job insecurity (United Nations Population Fund, 2025). Germany, Greece, Hungary, Italy, Poland and Spain subsidize or guarantee home loans for families, with terms that improve with the number of children (KfW, 2025; Hellenic Republic, 2026; Government of Hungary, 2023; CONSAP, 2026; Sejm of the Republic of Poland, 2024; Boletín Oficial del Estado, 2024); South Korea lends at subsidized rates to households with a child under two, to buy a home or pay a rental deposit (Ministry of Land, Infrastructure and Transport, 2023), and Singapore gives families priority for public flats (Ministry of National Development, 2026). In the 2024 US presidential campaign, Kamala Harris proposed up to $25,000 in down-payment assistance for first-time buyers (Harris for President, 2024), and JD Vance argued that young people are delaying families "because they can't afford a place to raise a family" (NewsNation, 2024).

[3] In the United States, 43% of owned homes have seven or more rooms, against 7.5% of rented homes (2023 American Housing Survey).

than when it only eases financial constraints. For example, removing the down payment for all households and expanding the housing supply raise births by about the same amount on impact, 1.3% and 1.4%. Yet, the gain from the credit easing policy is gone within four years, while the gain from the supply expansion keeps growing, to 3.0% by 2063. Credit available only to parents raises births by 5.1% on impact and by 16.6% by 2063.

The reason for these different effects is what each policy changes for a couple deciding whether to have a child. Lowering the down payment makes it easier to buy a home, but does not affect the flow cost of children. As this policy does not lower the burden of parenthood, and it does not change the demand for space, most of the households who use it buy the small homes they would otherwise have rented. As a comparison, increasing housing supply lowers prices and rents, and with them the cost of family space in the present and future. Credit offered only to parents does not lower the flow cost of children either, but it ties the gain to having a child: it raises the value of parenthood relative to childlessness, not only the value of owning.

These findings speak to a growing empirical literature that links housing costs and access to credit with births (Dettling and Kearney, 2014; Lovenheim and Mumford, 2013; Hacamo, 2021; Cumming and Dettling, 2024; Dettling and Kearney, 2026; van Doornik et al., 2024). My model is consistent with the positive effect of easier credit on births that these studies find, and it complements this evidence in two ways. First, a structural general-equilibrium model helps in disentangling the forces at play. By jointly modeling the households' choice of children, tenure, housing and savings, the model can shed light on the interaction between housing and children in equilibrium, and it can extrapolate to policies outside the range of existing evidence, such as credit available only to parents. Second, it allows a long-run comparison of policies. Empirical designs typically operate at short horizons and given prices. Some studies look at longer timeframes (Dettling and Kearney, 2026; van Doornik et al., 2024), but they are necessarily retrospective: it takes more than two decades for the effect of a policy on completed fertility to unfold. The model instead allows me to project forward. Starting from the economy of 2023, I follow each policy for several decades, as cohorts age and prices adjust.

I start by documenting how housing changes when a child arrives. Using the Panel Study of Income Dynamics, I show that three to four years after their first birth, parents live in 1.44 more rooms than two to three years before it, a 31% increase over 4.71 rooms, and they are 27 percentage points more likely to own their home, up from 43%. Moving to obtain more space becomes a third more likely after the birth, while moving for a better neighborhood does not.

I then build a life-cycle model in which households face uninsurable earnings risk. They save, decide whether to have children, whether to rent or own, and how much space to live in. A child requires a minimum amount of space while it lives at home. The largest homes are only available for sale, buying requires a down payment, and owners can borrow against their home. Housing is supplied along an upward-sloping supply curve, so house prices rise with the number of households, and therefore fertility choices feed back into the price of space a generation later.

To quantify the model, I approximate the United States economy of 2007 with a steady state in which births are at replacement. I assign some parameters from external evidence and choose the others so that this steady state matches fertility, housing and wealth moments of 2007. Holding these parameters fixed, I then let the preference for children fall after 2007 so that the model matches the decline in fertility through 2023, and I solve for the transition of the economy as prices, family composition and household formation adjust.

Finally, I study a range of policies, starting from the 2023 point in this transition. The results show why focusing on the immediate impact of a policy can be misleading. Credit for all buyers raises births immediately, but most of this effect disappears in four years, as many of the children born early would otherwise have been born later. Births then rise again, to 0.6% above the path without policy by 2063, as households that enter later gain more from credit while housing becomes cheaper. Instead, a supply expansion with the same effect on impact keeps raising births, to 3.0% above the path without policy by 2063, as each cohort that enters adulthood benefits from lower housing costs. Credit for parents raises births by 16.6% by 2063. In the long run, births per household return to replacement in every scenario, so the lasting gains of a policy show up as more households rather than more children per household: with credit for parents, there are 36% more households in the new steady state, and the house price is 33% higher. The main lesson is that housing policy aimed at births should be judged by its long-run effects, not only by its effect on impact.

1.1 Related work

This paper contributes to many strands of the literature on fertility, housing, and macroeconomics.

First, the paper relates to the large literature on housing and childbearing. This literature shows that fertility is lower where rent per room is higher (Simon and Tamura, 2009), dwelling size predicts first births (Ström, 2010), and higher house prices raise births among owners (Dettling and Kearney, 2014; Lovenheim and Mumford, 2013) but lower them among renters (Dettling and Kearney, 2014). These papers are mostly empirical; I study these effects in a general equilibrium model, which lets me focus on complex interactions between housing, credit and fertility and their long-run patterns. Structural work has also modeled fertility and housing choices, estimating dynamic models on micro data (Khorunzhina and Miller, 2022; Couillard, 2025). I add to this by placing similar choices in a life-cycle economy where house prices respond to the number of households, and follow policies over several decades.

Second, this paper relates to the literature on credit, housing and fertility. Quasi-experimental studies find that credit raises births: an expansion of home credit in the mid-2000s led young households to buy homes and have children (Hacamo, 2021), lower interest rates on home loans raise births among affected households (Cumming and Dettling, 2024), federally backed home loans raised births, and possibly completed fertility, during the baby boom (Dettling and Kearney, 2026), and winners of housing-credit lotteries in Brazil have more children, especially at ages 20 to 25 (van

Doornik et al., 2024). I add to this literature in two ways. First, I separate the timing of births from their number. In line with Kaplan et al. (2020) lower down payments lead young households to buy the small homes they would otherwise have rented. As a result, births rise on impact but mostly shift earlier in agents' lives, with a modest effect over the decades. Second, I compare credit with policies that lower the cost of space or tie credit to births.

Third, it relates to work on earnings risk and fertility. Because a child is an irreversible commitment, prudent parents who face more risk want fewer children (Fraser, 2001). Quantitative life-cycle models find that uninsurable earnings risk delays and reduces births (Sommer, 2016; Ejrnæs and Jørgensen, 2020; Nakakuni, 2026), and de la Croix and Pommeret (2021) show that risk gives the postponement of a first birth an option value. Quasi-experimental studies find that job displacement and job insecurity lower fertility (Del Bono et al., 2012; Clark and Lepinteur, 2022). I focus on modelling the housing component of such commitments. Children affect housing demand, and by modelling tenure choice and credit constraints I can study how the response of fertility to earnings risk is influenced by the housing market.

Fourth, it builds on quantitative models of housing over the life cycle. These models study tenure choice under earnings risk, down payments and transaction costs (Fisher and Gervais, 2011; Halket and Vasudev, 2014; Boar et al., 2022), how credit conditions and beliefs affect house prices, rents and ownership (Kaplan et al., 2020; Greenwald and Guren, 2025), why young households own less than they used to (Fisher and Gervais, 2011; Paz-Pardo, 2024), and the intergenerational allocation of housing (Coven et al., 2026). I contribute to this literature by endogenizing the fertility decision, to understand fertility and housing over the life cycle.

Finally, although this paper does not try to explain the decline in fertility, it relates to the broad literature on its causes and consequences. The literature on the causes has documented the importance of the cost and timing of children, bargaining within couples, changing attitudes toward parenthood, shifting priorities across cohorts and the compatibility of career and family (Jones et al., 2010; Doepke and Kindermann, 2019; Kearney et al., 2022; Doepke et al., 2023; Kim et al., 2024; Kearney and Levine, 2026; Fernández et al., 2026). Work on the consequences studies the efficiency and optimal rate of population growth, whether low fertility stabilizes, and the economics of a shrinking world (Golosov et al., 2007; Ball, 2026; Coffey et al., 2026; Fernández-Villaverde and Norrick, 2026). I take the decline as given, driven by an exogenous fall in the preference for children, and study how, for given preferences, housing shapes fertility decisions.

The rest of the paper is organized as follows. Section 2 presents the evidence on housing around the first birth. Section 3 describes the model, and Section 4 its quantification and the transition from 2007 to 2023. Section 5 describes household choices in the calibrated model, Section 6 studies housing policies, and Section 7 concludes.

2 Empirical Evidence

In this Section, I show some motivating evidence on housing changes around the first birth using the Panel Study of Income Dynamics (PSID). I begin by describing changes in housing at parenthood. Three to four years after the birth, parents live in 1.4 more rooms than two to three years before it, a 31% increase, and are 27 percentage points more likely to own their home, up from 43%. Both margins, space and ownership, are central to the model of Section 3, and they crucially inform the quantification in Section 4. I then document the increased salience of space among the reasons households give for moving. One to two years after the birth, the probability of moving for more space rises by 2.1 percentage points, a 33% increase over 6.4%, while moves for neighborhood reasons do not rise. Among households that move, the share citing more space rises from 12% to 27% three to four years after the birth.

2.1 Data and Design

The PSID has followed a representative sample of U.S. families and their descendants since 1968, every year until 1997 and every two years since. I use the waves from 1968 to 2023.

In this Section, I focus on four housing outcomes. "Rooms" is the number of rooms in the home. "Owner" equals one if the household owns its home. Households that moved since the previous interview are asked why they moved, and I use the first reason they give. "Moved for space" equals one if the household moved and its first reason is expansion of housing, such as more space, and zero otherwise. "Moved for neighborhood" is defined in the same way for neighborhood reasons, such as a better neighborhood or being closer to friends or relatives; this category also includes moves to attend school. Appendix B details the variables and the sample construction.

I measure how housing changes around the first birth, relative to two to three years before it and relative to adults who never have a child. I use this reference, rather than the year before the birth, because households may move ahead of a planned birth. Formally, let Y_{it} be an outcome of adult i in survey year t and E_i the year of the first birth. I estimate

$$Y_{it} = \alpha_i + \lambda_t + x_{it}'\gamma + \sum_g \sum_{k \neq -3/-2} \delta_{g,k} \, \mathbf{1}\{E_i = g\} \, \mathbf{1}\{t - E_i \in k\} + \varepsilon_{it}, \tag{2.1}$$

where α_i and λ_t are person and survey-year fixed effects, x_{it} contains indicators for age and education, g indexes first-birth years, and k indexes two-year windows of event time $t - E_i$. The coefficient $\delta_{g,k}$ is the change in Y in window k for adults whose first birth is in year g, relative to the omitted window.

The treated group consists of adults who have a first child and who, two to three years before the birth, were already the reference person or the spouse of a household, so that their housing in the

omitted window is that of their own household.[4] The control group consists of all adults who never have a child; they carry no event-time indicators. To account for effects that differ across cohorts of first births, I estimate equation (2.1) with the estimator of Sun and Abraham (2021).[5] I report

$$\hat{\beta}_k = \sum_g w_{g,k}\, \hat{\delta}_{g,k}, \tag{2.2}$$

where $w_{g,k}$ is the share of first-time parents observed in window k whose first birth is in year g. I refer to β_k as the housing response to the first birth in window k. For example, $\beta_{+3/+4}$ is the change in Y three to four years after the first birth, relative to two to three years before it and relative to the change among never-parents. Observations are weighted by the PSID longitudinal weight.

2.2 Housing Around the First Birth

Figure 1 shows the response of housing choices around the first birth. Panel (a) shows the path of housing size. Two to three years before the birth, parents live in 4.71 rooms. Rooms rise by 0.62 in the birth window (+13%) and by 1.44 three to four years after the birth (+31%), and rise more slowly afterwards, to 1.8 more rooms eleven or more years after the birth. In the windows before the reference, the coefficients are below a tenth of a room. The adjustment is large and concentrated around the birth: two-thirds of the long-run increase occurs by one to two years after it. This indicates that, although the analysis does not rule out anticipation, children acquire particular salience around the birth itself, and the demand for space rises abruptly.

Panel (b) shows the path of homeownership. Two to three years before the birth, 43% of parents own their home. Ownership rises by 16 percentage points in the birth window (+37%) and by 27 points three to four years after the birth (+63%).[6]

Figure 2 explores the reasons households give for moving. Panel (a) shows the probability of moving for more space, and panel (b) the probability of moving for neighborhood reasons. Two to three years before the birth, 6.4% of households move for more space and 3.0% for neighborhood reasons. One to two years after the birth, the probability of moving for more space is 2.1 percentage points higher (+33%), while the probability of moving for neighborhood reasons does not rise.[7] This highlights a strong role for concerns about size of the dwelling at childbirth.

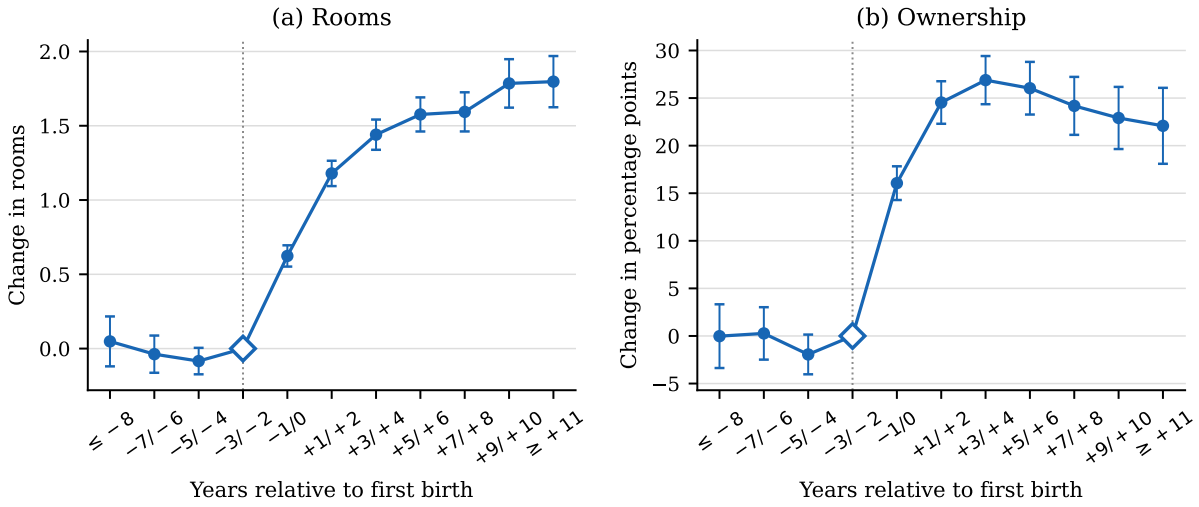
[4] The results are qualitatively similar when the sample includes, at each interview, every woman who is the reference person or the spouse of a household. For more detail, see Appendix B.4.

[5] Two-way fixed effects regressions with pooled event-time indicators are biased when the effect of the birth differs across cohorts (Goodman-Bacon, 2021). The method of Sun and Abraham (2021), followed here, estimates a separate path for each cohort of parents, compares it only with never-parents, and averages the paths with the cohort shares.

[6] Appendix B.4 also shows the probability of moving since the previous interview. It is highest two to three years before the birth, at 54%, and is 17 percentage points lower one to two years after the birth and 21 points lower three to four years after it (Table 8). Couillard (2025) finds the same anticipation of births in the ACS.

[7] Among households that move, the share citing more space rises from 12% to 27% three to four years after the birth (Table 8). A larger home could also come from a move to a cheaper location rather than from spending more on housing. Among households that already owned two to three years before the birth, house value rises by 30% three to four years after the birth and value per room by 11%, and moves across state lines become less frequent

Figure 1: Rooms and Ownership Around the First Birth (PSID)



Note: This figure shows the estimates of β_k in equation (2.2), in two-year windows, with 95% confidence intervals. Rooms use the sample on which the model's event study is measured ($N = 112{,}148$); ownership uses its available observations ($N = 111{,}272$). The hollow marker is the omitted reference window.

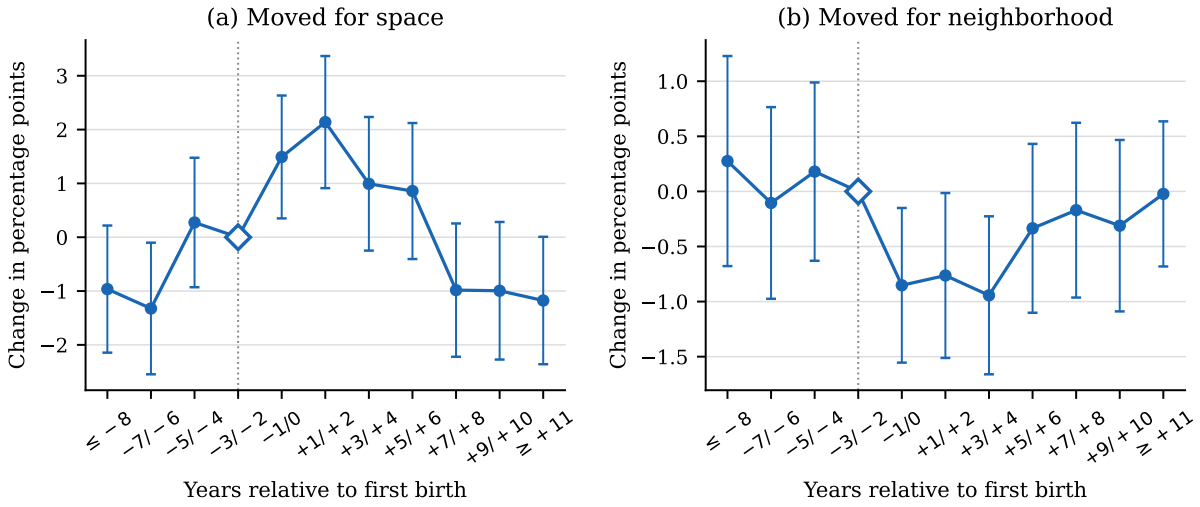
In summary, housing is particularly relevant around childbirth, which increases the desire for size and leads people to ownership. The complex web of interactions that take place in the data motivates the writing of a model. Agents need to make a set of intertwined decisions. Having children increases desire for space, and it is easier to get this space by owning (see Appendix A). Owning requires saving to be able to afford a down payment, and children also increase general expenses. The model developed in Section 3 tries to model the interaction between these forces, in order to discipline their relative strength and potential for policy. The evidence detailed above not only motivates, but also informs my modeling choices. Parents add about 1.4 rooms around the first birth: in the model, a child requires a minimum amount of space, which I measure by running the same event study in the model. Parents also become more likely to own. In the model, ownership gives access to larger homes, since the largest homes are only for sale. This is a potentially important channel, although Section 6.3 suggests it matters less than one might expect.

3 Model

In this Section, I develop a model of fertility and housing decisions. Households live a finite number of periods in an overlapping-generations economy. They face persistent idiosyncratic earnings risk and transitory taste shocks over fertility and housing. They derive utility from consumption, housing services and children at home, and they save in a one-period bond with an exogenous interest rate. They either rent or own their home. Owning gives more services than renting the same space, and

(Appendix B.5).

Figure 2: Reasons for Moving Around the First Birth (PSID)



Note: This figure shows the estimates of β_k in equation (2.2), in percentage points, with 95% confidence intervals ($N = 116{,}697$). The outcomes are probabilities among all households, including those that did not move. The hollow marker is the omitted reference window.

the largest homes are only for sale. Owners can borrow against their home, up to a fraction of its value, while renters cannot borrow. A buyer must therefore make a down payment, and a seller pays a transaction cost. During a fertile period of their lives, households choose whether to try for a child. Children raise the household's needs and the value of space, so the decision to have a child is also a decision about housing. Households work until retirement and then receive a pension, financed by a payroll tax on workers, who supply labor inelastically. House prices and rents are determined in equilibrium.

3.1 The economy

I begin by describing households and their preferences, the income process, housing, the assets available for trade and the fertility process. I then describe the housing market and firms. Time is discrete and indexed by t.

Households and Preferences. The economy is populated by overlapping generations of households, with adult age $a = 0, \ldots, J$. Households can have children until age a_F and work until age a_R, after which they retire. A household of age a survives to age $a+1$ with probability p_a. Figure 3 illustrates the different periods of the life cycle. Households differ in wealth, housing, labor efficiency, age and the number of children. During their fertile years, households can choose whether to try for a child. Throughout their life, they choose whether to rent or own their dwelling, how much housing to demand and how much to save.

Households value consumption c_t, housing services $\hat{h}_t$ and the number of children at home n_t, and

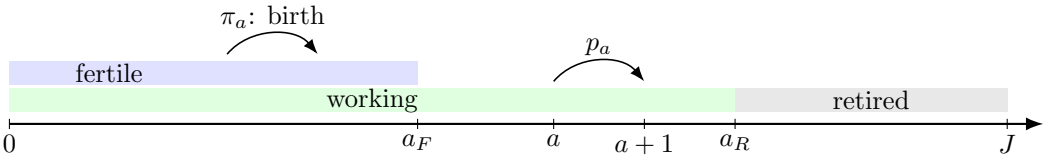


Figure 3: Life cycle of a household. Until age a_F, a try for a child succeeds with probability π_a, which declines with age. A household of age a survives to age $a+1$ with probability p_a.

discount the future by β. A household either owns h_t^O or rents h_t^R, and its housing services are

$$\hat{h}_t = \begin{cases} \chi\, h_t^O, & \text{owner}, \\ h_t^R, & \text{renter}, \end{cases} \qquad \chi > 1.$$

The parameter χ determines the additional services a household obtains from owning a home of a given size. Flow utility is

$$u(c_t, \hat{h}_t, n_t) = \frac{1}{1-\sigma}\left[\frac{c_t^{\alpha_0}\left(\hat{h}_t - \chi^{\mathbf{1}\{\text{own}\}}\,\bar{h}(n_t)\right)^{1-\alpha_0}}{e(n_t)}\right]^{1-\sigma} + \psi_t\, v(n_t), \tag{3.1}$$

with

$$\bar{h}(n_t) = h_P\,\mathbf{1}\{n_t > 0\}, \qquad v(n_t) = n_t^{1-\gamma}.$$

The first term is utility from a Cobb–Douglas composite of consumption and housing services above a requirement $\bar{h}(n_t)$, per adult equivalent, where α_0 is the weight on consumption and σ the curvature. Children raise the household's needs in two ways. First, a household with children at home must occupy h_P units of housing before additional housing yields services; an owner obtains $\chi(h_t^O - h_P)$ and a renter $h_t^R - h_P$. The requirement kicks in with the first child, and the household pays for it every period its children are at home, whatever the tenure choice. Second, the equivalence scale $e(n_t)$, with $e(0) = 1$, measures adult equivalents relative to a childless couple.[8] The second term, $\psi_t v(n_t)$, is the benefit of children at home: ψ_t is the preference for children and γ governs how fast the benefit of an additional child declines. The first child also carries a one-time utility cost ξ at birth.

Bequests. A household that dies at the end of period t leaves the estate $x_{t+1} = b_{t+1} + (1-\mu)P_t h_t^O$, its bond position and the value of its home net of the selling cost. As in De Nardi (2004), it values the estate according to

$$B(x_{t+1}) = \theta_0\,\frac{(\theta_1 + x_{t+1})^{1-\sigma}}{1-\sigma},$$

where θ_0 scales the strength of the bequest motive and θ_1 determines the extent to which bequests are a luxury good. Estates are collected by the government.

[8] I use the scale of Scholz et al. (2006), $e(n_t) = \big((2+\omega n_t)/2\big)^{\nu}$, where ω is the weight of a child relative to an adult and ν governs economies of scale.

Income. Before retirement, a household supplies one unit of labor inelastically, with efficiency $e_a z_t$. The term e_a is the age profile of earnings, and z_t is idiosyncratic labor efficiency, whose logarithm follows a first-order autoregressive process,

$$\log z_{t+1} = \rho_z \log z_t + \epsilon_{t+1}, \qquad \epsilon_{t+1} \sim N(0, \sigma_z^2),$$

where ρ_z is the persistence and σ_z^2 the variance of the innovations. Earnings are taxed at rate τ^w, and retirees receive a pension m_t:

$$y_t(a, z_t) = \begin{cases} (1 - \tau^w)\, w\, e_a z_t, & a < a_R, \\ m_t, & a \geq a_R. \end{cases}$$

Every household also receives a transfer T_t, the rebate of property-tax revenue.

Housing. There is one housing good, measured in units of space. Units can be combined and split at no cost, so all housing trades at one price P_t per unit and rents at one rent r_t per unit. Between periods, a fraction δ_H of every unit depreciates. A household enters period t with the home it owned at the end of the previous period, h_{t-1}^O, which is zero for a renter or a new household. Owner-occupied homes come in a finite set of sizes, $h_t^O \in \mathcal{H}$. Owners pay the property tax τ^p on the value of the home they choose, $P_t h_t^O$, and maintenance δ_H on the value of the home they enter the period with, $P_t h_{t-1}^O$, which restores it. Selling a home costs a fraction μ of its value. Renters choose any size up to a cap, $h_t^R \leq \bar{h}^R$, at rent r_t per unit, and reoptimize each period. Since $\bar{h}^R < \max \mathcal{H}$, the largest homes are only for sale.

Borrowing. Households save or borrow in a one-period bond, traded with the rest of the world at the exogenous gross return R_b. Let b_{t+1} denote the bond position a household chooses in period t. Borrowing must be secured by a home, so a renter cannot borrow, $b_{t+1} \geq 0$. An owner can borrow up to a fraction ϕ of the current value of its home:

$$b_{t+1} \geq -\phi\, P_t h_t^O. \tag{3.2}$$

The condition applies both to a household that buys a home and to one that remains in a previously owned home. A buyer must therefore pay at least $(1 - \phi) P_t h_t^O$ itself, out of its savings, the income of the period and the proceeds from selling the previously owned home if it had one.

Fertility. Until age a_F, a household decides each period whether to try to have a child. A try succeeds with probability π_a, which declines with age, and a success adds one child at home in the same period. Children make no decisions and affect the household by increasing housing and consumption needs and through the benefit in preferences. Each child at home leaves with probability λ per period, independently of its siblings, so a child stays at home for $1/\lambda$ periods on average. Children who leave form new households and enter the economy at age 0, with an initial

wealth position drawn from an exogenous distribution.

Construction. New housing is built by competitive developers from the consumption good. A developer that uses X_t units of the good builds I_t units of housing,

$$I_t = Z_t\, X_t, \qquad X_t \geq 0, \qquad Z_t = \Big(\frac{\bar{H}}{H_t}\Big)^{1/\eta}, \tag{3.3}$$

where Z_t is the number of units built per unit of the good. Productivity falls with the aggregate stock H_t, which stands in for land and regulation: each further unit is built on scarcer or more regulated land. The parameter $\bar{H} > 0$ sets the scale of the stock and $\eta > 0$ is the price elasticity of supply. Housing depreciates at the rate δ_H, and the stock evolves as

$$H_t = (1 - \delta_H)H_{t-1} + I_t. \tag{3.4}$$

Construction is irreversible, $I_t \geq 0$.

Rental companies. Rental housing is owned by rental companies, which live outside the economy. A rental company enters period t with the units it held at the end of $t-1$, of which $1-\delta_H$ per unit survive. It buys or sells units at P_t, in the same market as households, and rents all the units it holds at the end of the period to households, at the rent r_t per unit, paid at the start of the period. Rental companies pay the property tax $\tau^p P_t$ per unit, finance their purchases at the gross return R_b and, unlike households, pay no transaction cost when they sell.

Ownership of housing. At the end of each period, every unit of housing is owned either by the household that lives in it or by a rental company. Developers sell all they build in the period they build it. The home of a household that dies at the end of period t is part of its estate, valued at $(1-\mu)P_t h_t^O$; the government collects it and sells the units in period $t+1$. All sales take place in one market at the price P_t, where households, rental companies, developers and the government trade units.

Firms. A representative firm produces the consumption good, which is the numeraire, with a linear technology in effective labor,

$$Y_t = A\, L_t, \qquad L_t = \int_{a_t < a_R} e_{a_t} z_t \, \mathrm{d}G_t(\mathbf{s}_t, \boldsymbol{\varepsilon}_t),$$

where A is productivity, G_t is the measure of households defined in Section 3.4, and L_t is the total efficiency units of labor supplied by households of working age. The firm is competitive and pays each efficiency unit its marginal product, $w = A$. The wage is therefore constant over time. Developers use the good, not labor, so construction does not change earnings; the goods market clears through net exports.

Government. The government runs a pay-as-you-go pension system, taxes housing and collects estates. It taxes earnings at the constant rate τ^w and pays the same pension m_t to every retired household. The pension system balances its budget every period, so the pension equals payroll-tax revenue divided by the number of retired households. When the number of retirees rises relative to effective labor, the pension falls. The government also levies the property tax τ^p on the value of all housing, owned and rented, and returns the revenue to households as a lump-sum transfer T_t, equal across households. Finally, the government collects the estates of the households that die. It uses them to pay new households their initial wealth and spends the remainder on government consumption, from which households derive no utility.

3.2 The household's problem

I next characterize the household's problem recursively. A period has two stages, shown in Figure 4. At the beginning of the period, the household observes its fertility taste shocks and decides whether to try for a child. After the birth outcome, it observes its housing taste shocks and chooses its home and saving. A birth therefore changes the demand for housing in the same period. The state of a household is its bond position b_t, the home it owns h^O_{t-1}, its efficiency z_t, its age a_t and the number of children at home n_t.[9]

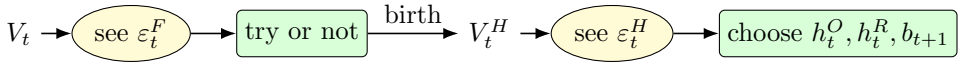


Figure 4: Timing within period t.

Fertility. Let $V_t(b_t, h^O_{t-1}, z_t, a_t, n_t)$ denote the value of a household at the beginning of period t, before it observes its fertility shocks. A household of fertile age chooses between not trying, $j = 0$, and trying, $j = A$. Each option receives a taste shock $\varepsilon^F_{t,j}$, i.i.d. Gumbel with scale κ and mean zero, with $\kappa = \kappa_1$ before the first child and $\kappa = \kappa_C$ after. The value of the household and the probability of trying are

$$\begin{aligned} V_t &= \mathbb{E}_{\varepsilon^F_{t,j}} \max_{j\in\{0,A\}} \left\{V^F_{t,j} + \varepsilon^F_{t,j}\right\} = \kappa \log\left[\exp\left(V^F_{t,0}/\kappa\right) + \exp\left(V^F_{t,A}/\kappa\right)\right], \\ \pi^F_{t,a} &= \frac{\exp\left(V^F_{t,A}/\kappa\right)}{\exp\left(V^F_{t,0}/\kappa\right) + \exp\left(V^F_{t,A}/\kappa\right)}, \end{aligned} \tag{3.5}$$

[9]The household also knows whether it has ever had a child: a birth is the first only if none was born before, whether or not that child is still at home. I keep this indicator out of the notation; it enters only ξ, κ_1 and the count of children ever born.

where $V^F_{t,0}$ and $V^F_{t,A}$ are the values of not trying and trying before the shocks. Both depend on V^H_t, the value of the household after the birth outcome, which I define below:

$$V^F_{t,0} = V^H_t(b_t, h^O_{t-1}, z_t, a_t, n_t),$$
$$V^F_{t,A} = \pi_a\big[V^H_t(b_t, h^O_{t-1}, z_t, a_t, n_t+1) - \xi\,\mathbf{1}\{\text{first child}\}\big] + (1-\pi_a)\,V^F_{t,0}.$$

Not trying leaves the number of children at home unchanged. Trying succeeds with probability π_a, adds a child at home and, for the first child, carries the utility cost ξ; if it fails, the household is in the same position as if it had not tried. Given its state, a household has a child with probability $\pi_a\pi^F_{t,a}$. Outside the fertile ages, there is no fertility choice and $V_t = V^H_t(b_t, h^O_{t-1}, z_t, a_t, n_t)$.

Housing and savings. After the birth outcome, the household observes its housing taste shocks and chooses its home, its tenure and its bond position. Let n denote the number of children at home after this period's birth outcome, n_t or n_t+1. Before the housing shocks, the value of the household is

$$V^H_t(b_t, h^O_{t-1}, z_t, a_t, n) = \mathbb{E}_{\varepsilon^H_{t,j}} \max_{h^O_t, h^R_t, b_{t+1}} \Big\{ u(c_t, \hat{h}_t, n) + \varepsilon^H_{t,j} + \beta\,\mathbb{E}_t\Big[p_a V_{t+1}(b_{t+1}, h^O_t, z_{t+1}, a_t+1, n_{t+1}) + (1-p_a)B(x_{t+1})\Big]\Big\}, \quad (3.6)$$

subject to the budget constraint of the chosen option,

$$\begin{aligned}
&\text{Buy:} && c_t + b_{t+1} + \tau^p P_t h^O_t + \delta_H P_t h^O_{t-1} = R_b b_t + y_t + T_t - P_t h^O_t + (1-\mu)P_t h^O_{t-1},\\
&\text{Stay:} && c_t + b_{t+1} + \tau^p P_t h^O_{t-1} + \delta_H P_t h^O_{t-1} = R_b b_t + y_t + T_t, \qquad (3.7)\\
&\text{Rent:} && c_t + b_{t+1} + r_t h^R_t + \delta_H P_t h^O_{t-1} = R_b b_t + y_t + T_t + (1-\mu)P_t h^O_{t-1},
\end{aligned}$$

and the borrowing limit (3.2) for buyers and stayers, or $b_{t+1} \geq 0$ for renters. A buyer sells the home it owns, if any, and buys a new one; a stayer keeps its home, $h^O_t = h^O_{t-1}$; a renter sells the home it owns, if any, and rents. Every owner pays maintenance $\delta_H P_t$ per unit on the home it enters the period with, which restores it, and a seller then receives $(1-\mu)P_t$ per unit. The expectation $\mathbb{E}_t$ is over next period's efficiency and the departure of children, which takes the household from n to n_{t+1} children at home. The shocks $\varepsilon^H_{t,j}$, i.i.d. Gumbel with scale κ_T and mean zero, are one per option j: renting, or owning each $h^O \in \mathcal{H}$. Let V^j_t denote the value of option j before the shock. Then

$$V^H_t = \kappa_T \log \sum_j \exp\big(V^j_t/\kappa_T\big), \qquad \pi^H_t(j) = \frac{\exp(V^j_t/\kappa_T)}{\sum_{j'}\exp(V^{j'}_t/\kappa_T)}, \quad (3.8)$$

where $\pi^H_t(j)$ is the probability of choosing option j. An option that leaves no saving choice within the budget and borrowing constraints cannot be chosen. Housing costs, credit and earnings risk shape V^H_t, and through it the gap $V^F_{t,A} - V^F_{t,0}$ that governs fertility.

Staying or selling. Owners are part of the supply of housing when they decide to sell. An owner that enters period t with the home $h^O_{t-1} > 0$ stays if

$$\underbrace{V_t^{h^O_{t-1}} + \varepsilon^H_{t,h^O_{t-1}}}_{\text{stay}} \;\geq\; \underbrace{\max\Big\{V_t^R + \varepsilon^H_{t,R},\; \max_{h \neq h^O_{t-1}} \big(V_t^h + \varepsilon^H_{t,h}\big)\Big\}}_{\text{sell, then rent or buy}},$$

and sells otherwise. The owner compares the value of keeping its home with the value of selling it and then renting or buying another, taking into account the future path of prices.

3.3 The problems of developers and rental companies

Developers. A developer takes the price P_t and productivity Z_t as given and solves

$$\max_{X_t \geq 0} \; P_t Z_t X_t - X_t.$$

With constant returns and free entry, developers earn no profit: they build only if a unit sells for what it costs, $P_t = 1/Z_t$, and build nothing if the price is below the cost of building on the stock that survives from the previous period.[10] By (3.3) and (3.4), the stock is then

$$H_t = \max\big\{(1-\delta_H)H_{t-1},\; \bar{H}\, P_t^{\eta}\big\}. \tag{3.9}$$

Rental companies. The value of a rental company that holds H^L_{t-1} units at the end of period $t-1$ is

$$J_t\big(H^L_{t-1}\big) = \max_{H^L_t \geq 0}\Big\{\big(r_t - \tau^p P_t\big)\, H^L_t - P_t\big[H^L_t - (1-\delta_H)H^L_{t-1}\big] + \frac{1}{R_b}\, J_{t+1}\big(H^L_t\big)\Big\}.$$

The first term is the rent net of the tax, the second the cost of net purchases and the third the continuation value. As a result, in equilibrium rental companies are indifferent between holding a unit of housing for one period and investing its price in the bond,

$$R_b\big(r_t - \tau^p P_t\big) + (1-\delta_H)P_{t+1} = R_b P_t,$$

that is,

$$r_t = (1+\tau^p)\, P_t - \frac{(1-\delta_H)\, P_{t+1}}{R_b}. \tag{3.10}$$

[10] Both the aggregate stock and the supply of housing to the market have a kink at the price at which nothing is built. The supply to the market, the left-hand side of (3.12), adds construction to the homes that owners sell and the homes of households that died. Above the kink, the stock rises with the price with elasticity η, and construction adds to the supply. Below it, developers stay out of the market, so the supply comes only from existing homes. The stock is thus elastic when demand rises and rigid when it falls, as in Glaeser and Gyourko (2005, 2018). If developers built in $t-1$, construction stops only if the price falls by more than $1-(1-\delta_H)^{1/\eta}$ in one period. In a steady state, construction replaces depreciation, and $H = \bar{H}P^{\eta}$.

Rental companies supply any quantity of rental housing at this rent. The rent equals the user cost of housing: the price and the property tax, net of the discounted value of what survives to the next period. When the price is expected to fall, the rent rises relative to the price, because rental companies must be compensated for the expected capital loss.

3.4 Equilibrium

Let G_t be the measure of households over pairs $(\mathbf{s}_t, \boldsymbol{\varepsilon}_t)$, where $\mathbf{s}_t = (b_t, h^O_{t-1}, z_t, a_t, n_t)$ is the state at the beginning of period t and $\boldsymbol{\varepsilon}_t$ collects the shocks drawn within it: the fertility shocks $\boldsymbol{\varepsilon}^F_t$, whether a try would succeed, which it does with probability π_a, and the housing shocks. Let N_t be the number of households. A household first chooses whether to try, $f_t(\mathbf{s}_t, \boldsymbol{\varepsilon}^F_t)$, and after the birth outcome chooses its home $h^O_t(\mathbf{s}_t, \boldsymbol{\varepsilon}_t)$, the space it rents $h^R_t(\mathbf{s}_t, \boldsymbol{\varepsilon}_t)$ and its bond position $b_{t+1}(\mathbf{s}_t, \boldsymbol{\varepsilon}_t)$.[11]

Definition 1. Given an initial measure of households G_0, initial stocks H_{-1} and H^L_{-1}, and the homes of the households that died at the end of period -1, an equilibrium is a path of prices $\{P_t, r_t\}$, stocks $\{H_t, H^L_t\}$, construction $\{I_t\}$, pensions $\{m_t\}$, rebates $\{T_t\}$, value and policy functions, and measures $\{G_t\}$ such that every period:

1. *Households* optimize: f_t, h^O_t, h^R_t and b_{t+1} solve the problem of Section 3.2, given prices, pensions and transfers.
2. *Developers* optimize: the stock satisfies (3.9).
3. *Rental companies* optimize: the rent satisfies (3.10).
4. The *rental market* clears at the rent r_t:

$$\underbrace{H^L_t}_{\text{held by rental companies}} = \underbrace{\int h^R_t(\mathbf{s}_t, \boldsymbol{\varepsilon}_t)\, \mathrm{d}G_t(\mathbf{s}_t, \boldsymbol{\varepsilon}_t)}_{\text{rented by households}}. \tag{3.11}$$

[11] Outside the fertile ages, $f_t = 0$. A renter has $h^O_t = 0$ and an owner $h^R_t = 0$.

5. The *owner market* clears at the price P_t, with construction $I_t = H_t - (1-\delta_H)H_{t-1}$:

$$\begin{aligned}
&\underbrace{I_t}_{\text{construction}} + \underbrace{\int \mathbf{1}\{h_t^O(\mathbf{s}_t,\boldsymbol{\varepsilon}_t) \neq h_{t-1}^O\}\,(1-\delta_H)h_{t-1}^O\,\mathrm{d}G_t(\mathbf{s}_t,\boldsymbol{\varepsilon}_t)}_{\text{homes of owners who sell}} \\
&+ \underbrace{\int (1-p_a)\,(1-\delta_H)h_{t-1}^O\,\mathrm{d}G_{t-1}}_{\text{homes of households that died in } t-1} \\
&= \underbrace{\int \mathbf{1}\{h_t^O(\mathbf{s}_t,\boldsymbol{\varepsilon}_t) \neq h_{t-1}^O\}\,h_t^O(\mathbf{s}_t,\boldsymbol{\varepsilon}_t)\,\mathrm{d}G_t(\mathbf{s}_t,\boldsymbol{\varepsilon}_t)}_{\text{homes bought}} + \underbrace{\delta_H \int \mathbf{1}\{h_t^O(\mathbf{s}_t,\boldsymbol{\varepsilon}_t) = h_{t-1}^O\}\,h_{t-1}^O\,\mathrm{d}G_t(\mathbf{s}_t,\boldsymbol{\varepsilon}_t)}_{\text{maintenance of owners who stay}} \\
&+ \underbrace{H_t^L - (1-\delta_H)H_{t-1}^L}_{\text{rental companies' net purchases}} .
\end{aligned} \tag{3.12}$$

6. The *goods market* clears: output equals consumption, the good used in construction, the costs of selling homes, government consumption and net exports (Appendix C).
7. *Pensions*: payroll taxes pay the pensions of the N_t^R retired households, $\tau^w w L_t = m_t N_t^R$.
8. *Rebate*: property-tax revenue is paid to all households, $T_t N_t = \tau^p P_t H_t$.
9. *Estates*: the government pays new households their initial wealth out of the estates it collects and consumes the remainder, C_t^g, given by (C.1).
10. *Population*: births are $B_t = \int \pi_a f_t(\mathbf{s}_t, \varepsilon_t^F)\,\mathrm{d}G_t(\mathbf{s}_t,\boldsymbol{\varepsilon}_t)$, and $G_{t+1} = \Gamma_t(G_t)$, where Γ_t takes the households that survive, with probability p_a, to the states $\mathbf{s}_{t+1} = (b_{t+1}, h_t^O, z_{t+1}, a_t+1, n_{t+1})$, in which n_{t+1} counts this period's births less the children who leave, and adds new households in proportion to past births.

Definition 2. A stationary equilibrium is a price P, a rent r, stocks H and H^L, construction I, a pension m, a rebate T, value and policy functions, and a measure of households G over $(\mathbf{s},\boldsymbol{\varepsilon})$, such that:

1. *Households* optimize, given P, r, m and T.
2. *Developers* optimize: by (3.9), the stock is $H = \bar{H}P^{\eta}$, and construction replaces depreciation, $I = \delta_H H$.
3. *Rental companies* optimize: by (3.10), the rent is $r = \big(1 + \tau^p - (1-\delta_H)/R_b\big)P$.
4. The *rental market* clears: $H^L = \int h^R(\mathbf{s},\boldsymbol{\varepsilon})\,\mathrm{d}G(\mathbf{s},\boldsymbol{\varepsilon})$.
5. The *owner market* clears: (3.12) holds with I, H^L and G constant. With the rental market, this means that owned plus rented housing equals the stock, $\int \big(h^O(\mathbf{s},\boldsymbol{\varepsilon}) + h^R(\mathbf{s},\boldsymbol{\varepsilon})\big)\,\mathrm{d}G(\mathbf{s},\boldsymbol{\varepsilon}) = H$.
6. The *goods market* clears (Appendix C), with constant aggregates.

7. *Pensions, rebate and estates*: $\tau^w wL = mN^R$, $TN = \tau^p PH$, and the government consumes the estates net of new households' initial wealth, C^g.

8. *The measure G is invariant*: $G = \Gamma(G)$.

3.5 Mechanisms

In this section, I describe three forces that shape fertility in the model. The first is timing: the probability of being successful at conceiving falls with age, while earnings and wealth rise, and owning a large home takes a down payment, which may weigh on fertility. The second is risk: the child's space is paid for in every period the child is at home, out of uncertain earnings. The third runs through the housing market: fewer children lower the demand for housing and its price, and a lower price raises fertility. I assess them quantitatively in Sections 4.4, 5 and 6.3.

Borrowing limits and the timing of children. The chance that a try succeeds, π_a, falls with age, while earnings $e_a z_t$ rise with age through e_a (Figure 5a). A child raises by $r_t h_P$ the spending needed for the same housing services, in every period it is at home. Homes larger than $\bar{h}^R$ are only for sale, and buying a home requires a down payment: a renter cannot borrow, and an owner can borrow only against its home, by (3.2). For a household whose earnings rise, a birth can make the borrowing limit bind harder, since the child raises its spending needs now, when its income is still low.[12] Such a household can delay the child, at the cost of a lower π_a. It can have the child in rented space and forgo the owner premium χ and the homes larger than $\bar{h}^R$. Or it can have the child and buy once its savings cover the down payment.

A higher loan-to-value ratio ϕ lowers the equity $(1-\phi)P_t h_t^O$ a buyer needs, so a household can buy earlier (Figure 5b), which may make it easier to have a child. The cost of the child's space, however, stays the same: by (3.10), that space costs about $r_t h_P$ every period, whether the household rents or owns.[13] For a change in ϕ in period t, (3.5) gives[14]

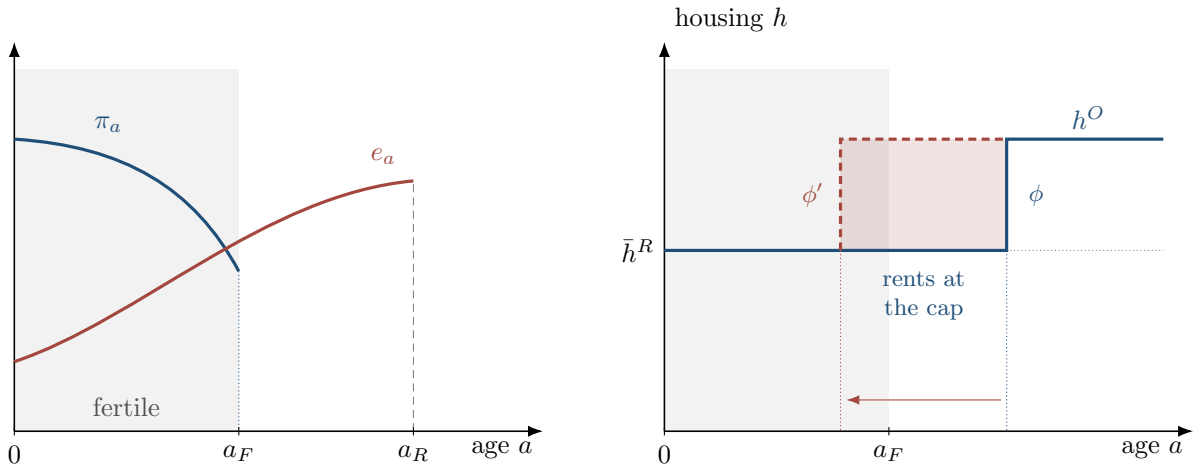
$$\frac{\partial \pi^F_{t,a}}{\partial \phi} = \frac{\pi^F_{t,a}\left(1-\pi^F_{t,a}\right)}{\kappa}\,\pi_a\left[\frac{\partial V_t^H(n_t+1)}{\partial \phi} - \frac{\partial V_t^H(n_t)}{\partial \phi}\right], \tag{3.13}$$

where $\partial V_t^H(n)/\partial\phi$ is the value of credit to a household with n children at home, positive only when the borrowing limit binds (Appendix D). Credit raises the probability of having a child only if it gives a particular benefit related to children: for example, if the household really cannot afford the down payment for a bigger house, and such a bigger house is the key element needed in order to have a kid. Section 6.3 shows that in the calibrated model credit mostly pushes young households to buy small homes they would otherwise have rented. Births come slightly earlier: completed fertility

[12] Appendix D derives the Euler equation for the bond and the conditions behind the results below.

[13] Apart from the selling cost, an owner that holds a unit from t to $t+1$ pays $(1+\tau^p)P_t$ in period t and holds $(1-\delta_H)P_{t+1}$ in period $t+1$, so by (3.10) its cost, in period-t units, equals the rent r_t.

[14] Here $\kappa = \kappa_1$ before the first child and κ_C after. A permanent change in ϕ adds the corresponding terms for all future periods, discounted and weighted by survival.



(a) Chance of a birth and earnings

(b) Housing by age

Figure 5: Timing of fertility, earnings and the purchase of a family home

Notes: Schematic. The shaded area marks the fertile ages. In panel (b), the solid line is the housing of a household that rents until its savings cover the down payment on a family home h^O at the loan-to-value ratio ϕ. The dashed line is the same household at $\phi' > \phi$, and the shaded area is the family space it gains.

falls slightly for the cohort aged 18 to 21 in 2023, and rises moderately for later cohorts.

Earnings risk. A child stays at home for $1/\lambda$ periods on average, so its space must be paid for in years of low earnings as well as high. Consider a renter at the borrowing limit, $b_t = b_{t+1} = 0$, that rents below the cap. Maximizing (3.1) subject to $c_t + r_t h_t^R = y_t + T_t$ gives the flow utility

$$u(c_t, h_t^R, n_t) = \frac{1}{1-\sigma}\left[\Theta_t \frac{y_t + T_t - r_t \bar{h}(n_t)}{e(n_t)}\right]^{1-\sigma} + \psi_t\, v(n_t), \tag{3.14}$$

where Θ_t is a function of the rent r_t.[15] A child enters the first term of (3.14) in two ways. Its space, $r_t h_P$, is a committed expenditure, paid out of income before the household divides the rest between consumption and additional space. The equivalence scale $e(n_t)$ then spreads what is left over more adult equivalents. The second term, $\psi_t v(n_t)$, is the value of the child. Let $\Delta_t(y_t)$ be the utility cost of a first child: the first term of (3.14) with $n_t = 0$ minus the same term with $n_t = 1$. Figure 6 shows two properties of this cost. First, it is higher when earnings are low, and convex in earnings, so a bad year raises the cost by more than a good year lowers it, and earnings risk raises its expected value. Second, this is stronger when the child's space is more expensive, so the lower rent of the new steady state also lowers the cost of risk. Both properties hold for $\sigma \geq 1$ (Appendix D). Savings soften this effect. Section 5.3 measures it in the calibrated model.

Households and the housing market. I refer to a stationary equilibrium (Section 3.4) as a steady state. In a steady state, the measure of households is invariant, so their number is constant:

[15] $\Theta_t = \alpha_0^{\alpha_0}(1-\alpha_0)^{1-\alpha_0} r_t^{\alpha_0 - 1}$.

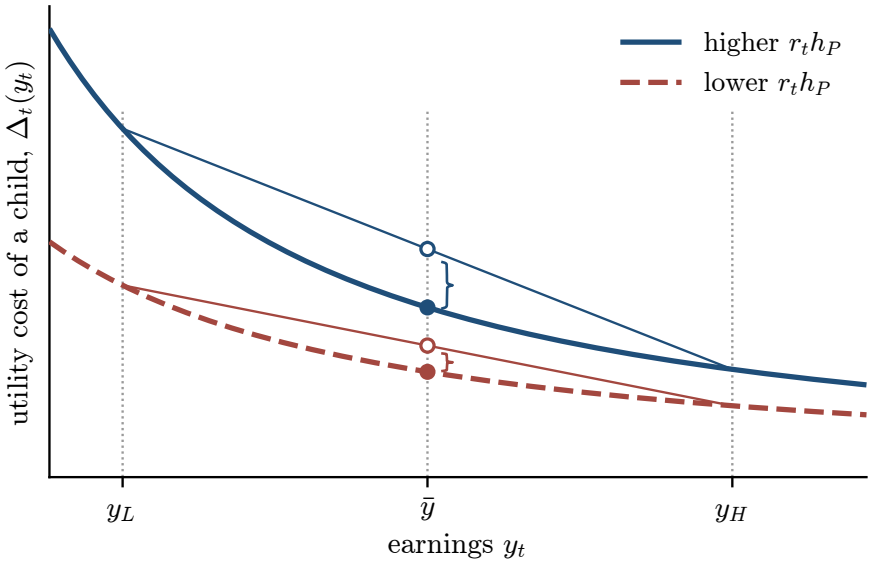


Figure 6: Earnings risk and the utility cost of a child

Notes: The figure is computed from (3.14) at the parameters of Section 4 (Appendix D). The curves plot the cost of a first child, Δ_t, against earnings for a renter with no savings, who spends its income each period, at the rent of the initial steady state (solid) and of the new steady state of Figure 7 (dashed). Earnings are y_L or y_H with equal probability. Filled circles mark the cost at expected earnings $\bar{y}$, open circles the expected cost, and the braces the additional cost due to risk.

each period, the children who leave home replace the households that die, and fertility is at the replacement rate $\bar{n}$. Since the household problem, the pension and the rebate per household do not depend on the number of households, the steady-state price is the one at which households choose replacement fertility, $n(P;\psi) = \bar{n}$. The housing market then determines the number of households N: the housing they own or rent, $N\,h(P;\psi)$, must equal the stock $\bar{H}P^{\eta}$, and construction replaces depreciation.[16]

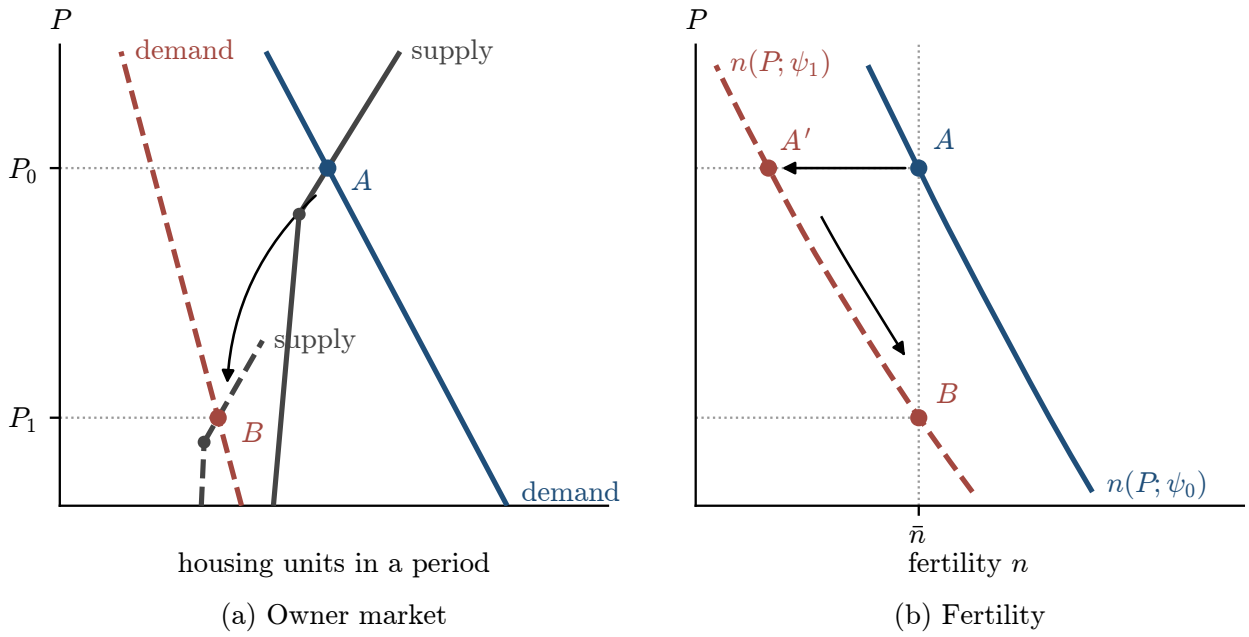
Figure 7 is helpful to understand how the housing market feeds back on fertility. It also illustrates the notion of steady state and the approach to the quantification in the next section. Starting from a steady state, the preference for children falls permanently from ψ_0 to ψ_1, and I compare the steady state the economy starts from with the one it converges to. Panel (b) plots the fertility households choose at each price, $n(P;\psi)$, under the two preferences. Panel (a) plots supply and demand in the market for owned homes (3.12), in each steady state. Supply comes from the homes of households that died, the homes of owners who sell, and construction, and has a kink at the price below which developers stop building.[17] Demand comes from households that buy a home, owners who stay and replace depreciation, and the net purchases of rental companies.

[16] Households respond to the rent. In a steady state, the rent is proportional to the price by (3.10), so housing and fertility can be read against the price. Along a transition, the rent also reflects the expected change in the price.

[17] Developers build only if the price covers the cost of adding to the homes that survive from the previous period. Above that price, construction makes supply elastic. Below it, nothing is built, and only existing homes are sold: the homes of owners who sell, which rise with the price, and the homes of households that died, which are sold at any price. Supply therefore never falls below the latter, hence the positive intercept. With fewer surviving homes, as at B, developers start building at a lower price.

A steady state is characterized by fertility being at replacement rate. The economy starts at A, where households choose replacement fertility at P_0 and the market clears with N_0 households. When the preference for children falls, households have fewer children at the same price, at A', so P_0 is no longer a steady-state price. Fertility is back at replacement when housing is cheaper, at P_1. At B, the market clears at that price with fewer households, N_1. Fewer homes are then bought and sold, so demand and the supply of existing homes are both smaller.

Between A and B, both curves move every period, as the supply depends on the homes that survive from the previous period and on who owns them. Section 4.4 computes this transition. Along it, smaller cohorts reduce the number of households and the demand for housing, and the price falls, but never by enough in one period to stop construction.



(a) Owner market (b) Fertility

Figure 7: The housing market and fertility

Notes: Panel (a) is a schematic of the owner market (3.12), drawn at the holdings of the initial steady state A (solid) and of the new steady state B (dashed). Dots mark the kinks of supply. Demand and the homes that come from existing owners are drawn in proportion to the number of households. Panel (b) is computed from solves of the model at given prices, at the parameters of Section 4 (Appendix D). It shows the fertility households choose at each price, under ψ_0 (solid) and ψ_1 (dashed), and the dotted line marks replacement. A' is the fertility households would choose under ψ_1 at the price P_0.

4 Quantification

In this section, I describe the quantification of the model. Because fertility rates vary a lot over time and the age distribution of the population reflects many years of past fertility decisions, some assumptions are needed to make the quantification tractable. To this end, I approximate the United States economy of 2007 with a steady state in which births are at replacement. I parameterize the model so that this steady state matches fertility, housing and wealth moments of 2007. Holding

these parameters fixed, I then let the preference for children ψ_t fall after 2007 so that the model matches the decline in fertility until 2023, and I solve for the transition of the economy as prices, family composition and household entry adjust (Section 4.4). I treat this fall as an exogenous shock to preferences and take it as given. The 2023 state of this transition is the starting point of the policy experiments in Section 6.

4.1 Quantification strategy

Two measures of fertility guide the quantification (Figure 8). Period fertility, or the total fertility rate, is the number of children a woman would have if she lived through her fertile years at the birth rates of a given year. Completed fertility is the number of children a woman has had by the end of her fertile years. I measure it at ages 40 to 44. Whether a population replaces itself depends on completed fertility, which is then the key number to look at when thinking about population growth trends. In a steady state the two measures coincide. Out of a steady state, completed fertility lags period fertility, because it adds up the births of a cohort over two decades. In the United States, period fertility peaked at about 2.1 in 2007 and fell to 1.62 by 2023, while completed fertility was still 1.92 in 2024. If period fertility stayed at its 2023 level, completed fertility would fall to 1.62 within about two decades, as the cohorts that had their children before 2007 age out.[18]

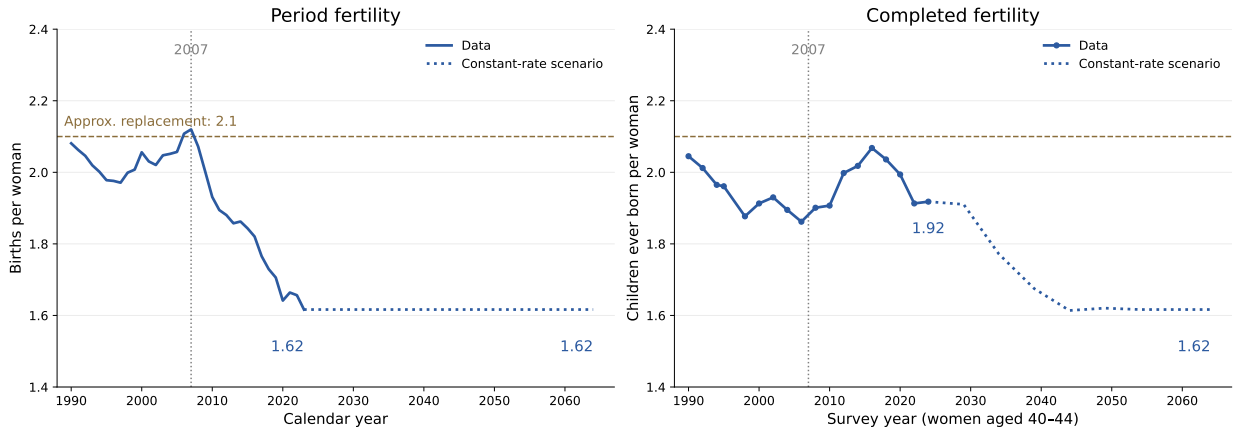


Figure 8: Period and completed fertility in the United States

Notes: Left: period total fertility rate, World Development Indicators. Right: children ever born to women aged 40–44, CPS Fertility Supplement. Dotted lines: a constant-rate scenario that holds period fertility at its 2023 level; the completed-fertility continuation is rescaled to end at that rate.

The transition follows the logic of Section 3.5: fertility falls first at about the inherited price, and the price falls as smaller cohorts enter. I compare each policy experiment with the path the economy follows from 2023 without it.

[18] The same lag then is the reason the number of households does not fall instantly when fertility falls: fewer births lower the number of new households only slowly, as the smaller cohorts reach adulthood. This is also why, despite almost two decades of below-replacement period fertility, the population of the United States has not yet started to decrease.

I divide the parameters of the 2007 economy into two groups. The first group is assigned from external evidence or measured directly in the data. The second is chosen so that the model matches a set of moments in the data. Table 2 reports the moments, and Table 1 the parameters.

4.2 Assigned parameters

Table 1: Parameter values

Assigned			Calibrated		
σ	2	risk aversion	β_{ann}	0.936	annual discount factor
(ω, ν)	0.7, 0.7	equivalence scale	ψ_0	0.177	preference for children
α_0	0.733	consumption weight	γ	0.646	curvature, benefit of children
λ	2/9	child departure, per period	ξ	0.295	first-birth utility cost
π_a	0.98–0.50	success of a birth attempt	κ_1	0.050	first-birth taste scale
ϕ	0.80	financed share	κ_C	0.129	later-birth taste scale
μ	0.06	selling cost	h_P	0.89	space requirement (rooms)
δ_H	0.014	depreciation, annual	χ	1.112	owner service multiplier
τ^p	0.0106	property tax, annual	κ_T	0.013	tenure taste scale
r	0.02	real interest rate, annual	θ_0	0.244	bequest-motive scale
τ^w	0.08	payroll tax	$\bar{H}$	6.776	housing supply scale
η	0.63	housing-supply elasticity			
$\bar{h}^R$	6	rental size cap (rooms)			
θ_1	0.008	bequest shifter			
(ρ_z, σ_z)	0.735, 0.484	earnings, per four years			

Notes: Assigned parameters are set outside the calibration (Section 4.2); calibrated parameters are chosen to match the moments of Table 2 (Section 4.3).

Demographics and timing. A period in the model lasts four years. Households enter at 18, retire at 66 and live at most through 85. Survival is certain before retirement and, after it, follows the 2023 period life table of the Social Security Administration. Births are possible through age 45. The probability that an attempt succeeds within a period falls with age as $\pi_a = 1 - 0.02\, e^{0.134(a-18)}$, and is zero from age 45, using the exponential form of Sommer (2016). The schedule gives 0.90 at 30, 0.80 at 35 and 0.62 at 40, close to the probabilities of a conception leading to a live birth within four years computed by Leridon (2004), 0.91, 0.84 and 0.64. A child leaves home with probability $\lambda = 2/9$ per period, so children stay at home for 18 years on average.[19]

Preferences. Flow utility (3.1) is

$$u(c, \hat{h}, n) = \frac{1}{1-\sigma}\left[\frac{c^{\alpha_0}\left(\hat{h} - \chi^{\mathbf{1}\{\text{own}\}}\,\bar{h}(n)\right)^{1-\alpha_0}}{e(n)}\right]^{1-\sigma} + \psi\, n^{1-\gamma}, \qquad e(n) = \left(\frac{2+\omega n}{2}\right)^{\nu}.$$

[19] Stocks, such as wealth and house prices, are in annual units, and flows, such as income, consumption and rent, accrue over the four years. Interest, discounting and depreciation compound over the period, so $R_b = 1.02^4$ and $\beta = \beta_{\text{ann}}^4$, and the property tax is four times its annual rate. A data statistic for an age window weights each four-year model cell by the share of the cell inside the window.

I set the curvature to $\sigma = 2$, a common choice in the literature. The equivalence scale $e(n)$ is that of Scholz et al. (2006), with $\omega = \nu = 0.7$: a couple with two children needs 1.45 times the consumption, and the housing services above its requirement, of a childless couple to reach the same utility. The consumption weight $\alpha_0 = 0.733$ is set to match the share of housing in the spending of childless renters in the 2019–2023 CEX, 26.7%. Indeed, a childless renter has no space requirement and no owner multiplier, so its utility is Cobb–Douglas in consumption and rented space, and it spends a share $1 - \alpha_0$ of its budget on rent.

Housing finance and taxes. The annual real interest rate is 2%, and selling a home costs 6% of its value, as in Greaney et al. (2025). Buyers can finance at most $\phi = 0.80$ of the price, the conventional limit above which a mortgage requires private insurance, and there is no limit on payments relative to income. I measure depreciation and the property tax at the national level, with the method of Greaney et al. (2025). Depreciation, 1.4% a year, is the 2007 depreciation of owner-occupied residential structures in the BEA fixed-asset accounts, divided by the value of the housing stock, land included, in the Financial Accounts of the Federal Reserve. The property tax, 1.06% of value, is the ratio of aggregate property taxes to the aggregate value of owner-occupied homes in the 2007–2011 ACS. Retirees receive a common pension equal to 23% of mean earnings, the ratio in the CPS ASEC, and the payroll tax that finances it is 8%.

Earnings. A household of age a earns $e_a z_t$, where

$$\log z_{t+1} = \rho_z \log z_t + \epsilon_{t+1}, \qquad \epsilon_{t+1} \sim N(0, \sigma_z^2).$$

I estimate ρ_z and σ_z from the variance and first autocovariance of log household earnings, averaged over non-overlapping four-year spells and net of age and year effects, in the 1984–1997 PSID, the last years in which the survey is annual. This gives $\rho_z = 0.735$ and $\sigma_z = 0.484$, which I approximate with nine states. The age profile e_a is hump-shaped and normalized to average one over working ages: earnings rise by about half from ages 18–25 to their peak at ages 34–45, and decline modestly before retirement.

Housing. Owners choose among homes of 2, 4, 6, 8 and 10 rooms, and renters any size up to $\bar{h}^R = 6$ rooms. The cap reflects the rental stock: in the 2023 AHS, 43% of owner-occupied homes have seven or more rooms, against 7.5% of rented homes (Appendix A). The supply elasticity in (3.9) is $\eta = 0.63$, from the estimates of floor-space supply of Baum-Snow and Han (2024).

Entry wealth. Households enter the economy with the wealth of young adults in the PSID: the net worth of childless renters aged 18 to 24. Negative values are set to zero, since households in the model cannot borrow without collateral.

Bequests. In the bequest motive, θ_1 determines how far bequests are a luxury good. I set it to 1% of median annual earnings in the model, $\theta_1 = 0.008$, as in Greaney et al. (2025).

4.3 Calibrated parameters

I choose the remaining eleven parameters so that the 2007 steady state matches the eleven moments in Table 2.[20] The parameters are chosen jointly, but some moments are more informative about some parameters than others. I group them into four blocks and, for each, describe the variation in the data that disciplines it and the mechanism in the model through which it does.

As the model contains many contemporaneous interactions, there are several things to keep in mind to properly inform the different parameters. First, fertility and housing are determined together. In the 2007 steady state, the house price is the one at which completed fertility equals replacement, so a parameter that lowers births also lowers the price, and a lower price raises the space households occupy and how often they own. I separate the two blocks with moments that respond to the price very differently. How births are spread across women and over the life cycle, measured by the shares of childless women and of mothers with one child and by the mean age at first birth, barely moves with the price. Mean rooms and ownership respond strongly to it. Mean rooms then pin the price, and the fertility moments identify the parameters that govern who has children and when. Second, households with children occupy more space than other households, partly because a child requires space and partly because parents are older and have higher earnings and savings. The change in rooms within households at the first birth, relative to adults who never have a child, isolates the space a child requires, h_P. The difference in ownership between recent parents and other households measures how closely tenure follows these needs, κ_T. Third, a change in when women have children and a change in how many they have both move period fertility. The distribution of completed family size at ages 40 to 44 measures how many children women have, and the mean age at first birth measures when they start.

Fertility. I choose the first-birth cost $\xi = 0.295$ to match the share of childless women aged 40 to 44 in the June Fertility Supplement of the CPS, pooling 2004 and 2006, 19.8%, and the curvature $\gamma = 0.646$ to match the share of mothers with exactly one child, 21.4%. Given that the cost is paid only at the first birth, a higher ξ raises childlessness, and mothers have larger families. A higher γ makes the benefit of an additional child decline faster, so childlessness falls and more mothers stop at one.

Timing of births. I include taste shocks in the fertility decision because, in the data, otherwise similar households make different choices. First births are spread over the whole fertile window, from the late teens to the early forties, and women in the lowest third of family income have about as

[20] The parameters minimize a sum of squared gaps between model and data moments, each divided by a tolerance for the moment (Appendix E).

many children by ages 40 to 44 as women in the top third. The shocks help explain this idiosyncratic variation. The scale of the first-birth shocks, $\kappa_1 = 0.050$, governs how much first births depart from the trade-off between fecundity and earnings of Section 3.5. A larger κ_1 spreads first births over the fertile window, so the mean age at first birth, 26.0 years in the NCHS natality files for 2003–2006, rises. I choose κ_1 to match it. The estimate is small: first births follow the economic trade-off closely, and the shocks mostly spread them across ages. The scale for later births, $\kappa_C = 0.129$, determines how willing mothers are to have more than one child. Since the option of further births raises the value of becoming a parent, a larger κ_C also brings first births forward, and I choose it to match the share of women who are mothers by age 25, 45.7% in the CPS.

Preference for children, housing and tenure. The preference ψ_0 is the taste for children in the 2007 steady state, before the decline of Section 4.4. I choose it so that completed fertility equals replacement, 2.1, and the supply scale $\bar{H}$ so that mean rooms are 5.73, as in the 2007 AHS. The two are chosen together: a larger $\bar{H}$ lowers the house price, households occupy more space, and replacement then requires a lower ψ_0. This gives $\psi_0 = 0.177$ and $\bar{H} = 6.776$.

I choose the owner multiplier $\chi = 1.112$ to match ownership among heads aged 30 to 55 in the 2005–2006 ACS, 67.6%.[21] An owner obtains χ times the services of the same home, so a higher χ raises the value of owning relative to renting.

The space requirement $h_P = 0.89$ rooms is disciplined by the change in housing within households at the first birth. Recall from Section 2.2 that future parents in the PSID live in 4.71 rooms two to three years before the first birth, and three to four years after it occupy 1.44 more rooms, relative to the change among adults who never have a child. I run the same regression on households in the model. The requirement is the space a household must add once it has a child, and this jump is the most direct measure of it in the data.

The tenure taste scale $\kappa_T = 0.013$ is disciplined by the difference in ownership between recent parents and other households. In the ACS, heads whose oldest child is under four own at a rate 12.8 percentage points higher than heads with no children at home. In the model, homes larger than $\bar{h}^R$ rooms are only for sale, so the space requirement pushes parents into ownership. A smaller κ_T makes tenure follow these needs more closely and widens the gap.

Wealth and bequests. I choose the discount factor $\beta = 0.936$ (annual) to match aggregate net worth of 4.46 times aggregate annual gross labor earnings in the PSID:[22] as more patient households save more, wealth accumulation responds directly to the discount factor. The bequest scale $\theta_0 = 0.244$ governs the incentives to hold on to wealth in old age. I choose it to match the annual flow of estates to children in the 2007 SCF, 0.73% of aggregate net worth.

[21] Both ownership samples exclude mobile homes, boats and vans (ACS structure codes 3 to 10), which have no counterpart in the model.

[22] Net worth of heads aged 18 to 85, excluding business and farm equity and other real estate; gross labor earnings of heads aged 18 to 65.

Table 2 shows the targets and the estimated moments.

Table 2: Moments used in the calibration

Moment	Data	Model
Completed fertility	2.10	2.10
Childless, women 40–44 (%)	19.8	20.1
Exactly one child, mothers 40–44 (%)	21.4	21.7
Mean age at first birth (years)	25.98	25.97
Mothers by age 25 (%)	45.7	45.5
Mean occupied rooms	5.73	5.75
Ownership, heads 30–55 (%)	67.6	66.9
Recent-parent ownership gap	0.128	0.125
First-birth room response	1.440	1.475
Wealth / annual gross earnings	4.46	4.35
Annual bequests / wealth (%)	0.73	0.73

Notes: Fertility: CPS June Fertility Supplement 2004 and 2006, women 40–44, and mothers by 25; NCHS natality 2003–2006 for the mean age at first birth; completed fertility is a normalization. Housing: AHS 2007 for rooms; ACS 2005–2006 for ownership and the gap between heads whose oldest child is under four and heads without children; PSID event study for the room response, measured in the model with the same design on simulated households. Wealth: PSID 2005 and 2007; bequests: SCF 2007. Owners choose among homes of 2, 4, 6, 8 and 10 rooms. The 2007 house price is 0.771.

4.4 The fertility decline, 2007–2023

Starting from the 2007 steady state, I hold all estimated parameters fixed and let the preference for children decline permanently from its 2007 level ψ_0 toward a lower level ψ',

$$\psi_t = \psi' + (\psi_0 - \psi')\,\rho^{\tau}, \tag{4.1}$$

where τ counts four-year periods since 2007. Households learn the whole path in 2007 and have perfect foresight about it and about prices. The house price, the pension and the rebate clear their markets at every date (Appendix F). I set $\rho = 0.35$ and choose $\psi' = 0.148$ to match period fertility in 2020–2023, 1.646 (Figure 9).[23] The decline in ψ_t summarizes changes in the incentives to have children that the model does not otherwise capture.[24] From the period dated 2023 on, period fertility is 1.63 and rises slowly, to 1.73 by 2063.

Figure 10 shows the whole transition. Period fertility returns to replacement slowly: it is 2.09 by 2203, and by 2403 the economy is at its new steady state, with the number of households at 49%, total housing at 67% and the house price at 53% of their 2007 levels. As the population shrinks, the house price falls: it is 3.6% below its 2007 level in 2023 and 19.3% below in 2063. The rent does not follow the price down at first. Along a transition, rent equals the user cost net of the expected

[23] A model period is dated by its first year and compared with the average period fertility of the four following calendar years: the period dated 2019 with 2020–2023.

[24] See Kearney et al. (2022) and Doepke et al. (2023) for reviews of the recent decline in fertility and its possible causes.

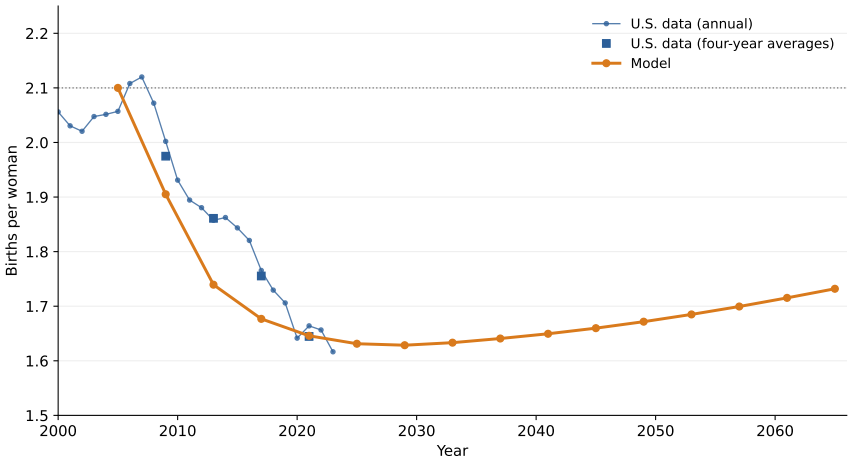


Figure 9: Period fertility, model and data, 2007–2063

Notes: Model: births per woman in each four-year period of the transition from the 2007 steady state. Data: annual period fertility and its four-year averages.

capital gain, so a price that is expected to keep falling raises the rent: in 2023 the rent per room is 2.0% above its 2007 level.

To illustrate the role of the housing stock in determining fertility in the model, the dashed lines in Figure 10 show the same transition with the total housing stock H held at its 2023 level from 2023 on. With a fixed stock, the price falls much further, to 47% of its 2007 level around 2111. At that price fertility overshoots replacement, reaching 2.18 around 2119, and the additional children fill the fixed stock a generation later. As they do, the price recovers and fertility returns to replacement. In the long run the price is the same as in the baseline, 53% of its 2007 level, because fertility at replacement pins it. The stock determines only how many households live at that price, 71% of the 2007 level instead of 49%. This suggests a potential role for housing supply in speeding up the recovery of fertility, which I explore in Section 6.

4.5 Testing the model

I conclude by comparing the model with some moments that the calibration does not target, reported in Table 3. The first is how many children mothers have beyond the first. This matters because the space requirement does not rise with the number of children, so later births test the other costs and benefits of children in the model. The calibration targets the share of women with no children and the share of mothers with one child, but not how the remaining mothers split between two children and three or more. The model matches this split: 32.2% of women aged 40 to 44 have two children and 30.4% have three or more, against 34.4% and 28.6% in the CPS. The second is the timing of first births, which matters because the central trade-off in the model is one of timing, between fecundity that falls with age and earnings and savings that rise with it. The model matches the share of first births at 30 or later, 24.2% against 24.9% in the natality files. The third is the

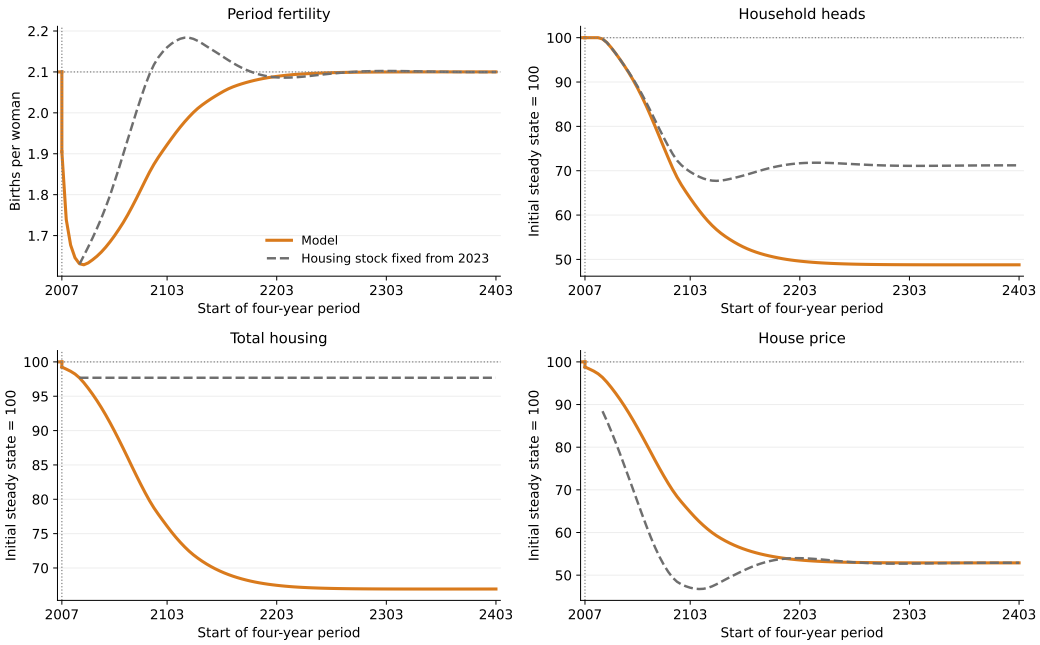


Figure 10: The full transition, 2007–2403

Notes: Period fertility, number of household heads, total housing and the house price along the transition, with the initial steady state normalized to 100. Dashed: the same transition with the housing stock held at its 2023 level from 2023 on.

dispersion of wealth among the old, which reflects how much households save: the ratio of the 90th percentile to the median of wealth relative to income at ages 76 to 84 is 3.27 in the model and 3.52 in the PSID. The model fits two housing moments less well. Children are linked to housing in the model through the space they require and the ownership of large homes. Households with three or more children occupy 0.57 more rooms than those with one or two, against 0.39 in the data, and ownership rises around the first birth by 14 percentage points, against 26.

Table 3: Moments not targeted, 2007

Moment	Source	Data	Model
First births at age 30+ (%)	NCHS 2003–2006	24.9	24.2
Rooms: 3+ versus 1–2 resident children	ACS 2005–2006	0.385	0.574
First-birth ownership response	PSID event study	0.261	0.141
Wealth/income p90 / median, ages 76–84	PSID	3.52	3.27

Notes: This table reports moments that do not enter the loss (E.1). The ownership response is the event-study coefficient of Section 2 on the model's observation support, measured on simulated households with the same design.

Figure 11 compares the life-cycle profiles of ownership, rooms and first births by age with the data. The calibration only targets ownership among heads aged 30 to 55, mean rooms and the mean age at first birth, so the shape of these profiles is untargeted. The model reproduces fairly well the rise of ownership with age. It also matches the hump in housing size: rooms rise until the forties and

then decline, although the model profile is steeper. First births by age of the mother also follow the data closely. Young households own too often in the model, 34% at 18–21 and 44% at 22–25, against 12% and 24% in the data, and older households keep their homes, since the model has no health or long-term-care risk that would lead them to sell.

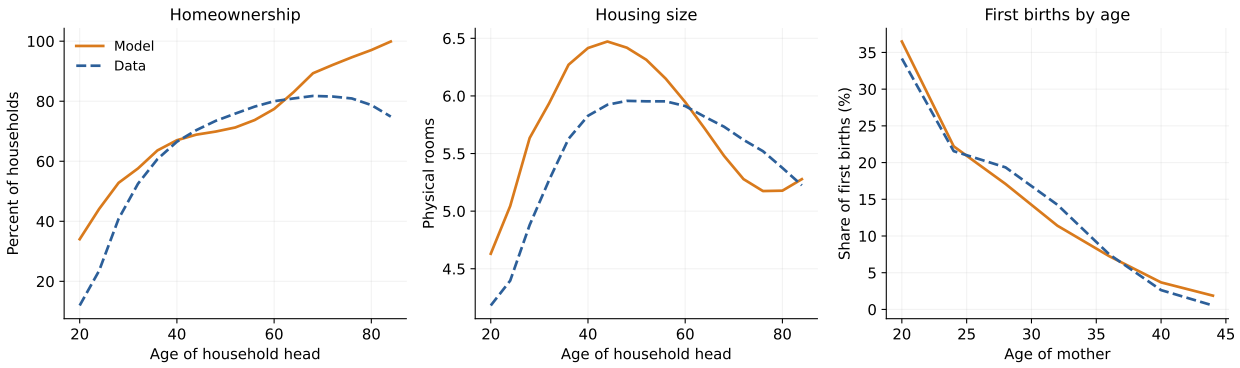


Figure 11: Life-cycle profiles in 2007, model and data

Notes: Homeownership and rooms by age of the household head, and the distribution of first births by age of the mother. Model: 2007 steady state. Data: ownership and rooms (at most nine) from the 2005–2006 ACS; first births by four-year age group from the NCHS natality files for 2005–2007.

Appendix G compares the 2023 economy, the starting point of the policy experiments, with the 2023 data. None of these moments is targeted by the transition. The model reproduces the direction of the shift in the timing of births: from 2007 to 2023 the share of women who are mothers by 25 falls by 13 percentage points in the model and by 18 in the data, in line with the fall in birth rates at younger ages documented by Kearney et al. (2022). It also matches the share of mothers of only one child, 22.2% in both, and mean rooms, with the model producing 5.51 against 5.58 in the data. To illustrate a shortcoming of modelling the decline through a single preference shock, take childlessness. In the model, the decline operates similarly on every cohort alive, so childlessness among women aged 40 to 44 rises to 28.7%. In the data, instead, it moved only from 19.8% to 18.8%.

5 Household Choices in the Calibrated Model

In this section, I explore how households in the calibrated model respond to parenthood, and how earnings risk shapes the decision to have kids. First, I show that having children changes housing, consumption and tenure. A child increases the demand for space, and the cost weighs most on households with little wealth or low earnings. Second, I illustrate the impact of income risk: lowering the standard deviation of earnings shocks raises completed fertility.

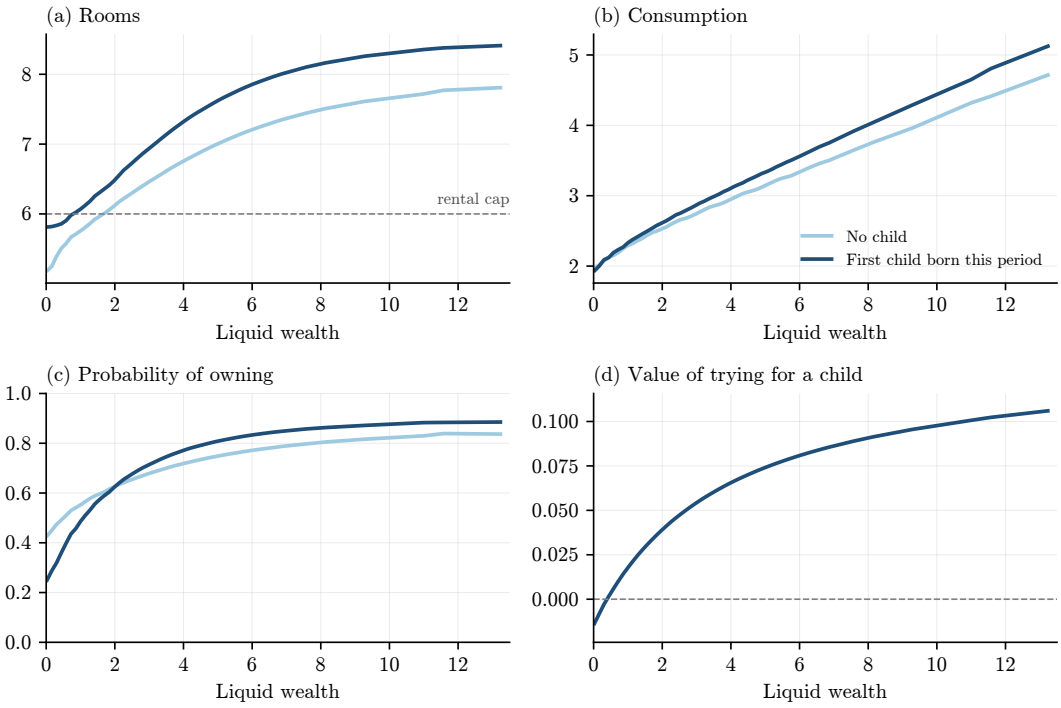


Figure 12: Choices with and without a first child

Notes: Renter aged 26–29 with median earnings ($z = 0.78$), 2007 stationary equilibrium. Liquid wealth in units of mean annual earnings. Rooms and consumption are averages over tenure choices.

5.1 Choices at the first birth

Figure 12 compares the choices of the same household in the period in which it has no child and in the period in which a first child is born. The household is a renter aged 26 to 29 with median earnings, and the figure varies its liquid wealth. Without savings, a first child pushes the space the household rents close to the rental cap and lowers the probability of owning from 42% to 24%. The child needs space when the household can least afford a down payment, so it rents and pays for the space out of consumption. With more than about two years of earnings in liquid wealth, the child instead raises the probability of owning, and the household buys a larger home.

Panel (d) shows the value of trying for a child, $V^F_{t,A} - V^F_{t,0}$. It is negative without savings and rises steeply at low wealth before flattening. Wealth therefore matters for the decision to have a first child, disproportionately so among poor households.

5.2 Choices across earnings

Figure 13 repeats the comparison across earnings states, for renters aged 26 to 29 without savings. A first child adds about two thirds of a room at earnings up to the median, and about half a room at high earnings, where most of these households buy a home even without a child. Consumption falls by 21% at the lowest state and by 2% at the median state. At high earnings it rises, because

the equivalence scale raises the household's needs and these households can afford to meet them. As discussed in Section 3.5, the space requirement does not depend on renting or owning, so owning does not avoid its cost. Since this cost is the same for every household, it weighs most on those with low earnings.

5.3 Earnings risk

Section 3.5 showed that earnings risk raises the expected cost of a child, since that cost falls with earnings at a decreasing rate. To measure this force, I lower the standard deviation of the four-year earnings shock by about a quarter, from 0.48 to 0.35, rescale the earnings states to keep mean earnings unchanged, and solve the model at the 2007 price. Lower risk raises completed fertility by 12%, from 2.10 to 2.36. Figure 15 below reports the effect over a range of values of earnings risk.

Figure 14 shows where the additional births come from. In the baseline, the model matches the distribution of children ever born among women in their early forties. Lower risk shifts households out of childlessness and toward larger families: the share without children falls from 20% to 12%, and the shares with two and with three or more rise.

Earnings risk weighs more when a child requires more space. Figure 15 solves the model at the 2007 price for different values of the space a child requires, h_P, and of the standard deviation of earnings shocks. Lowering risk to 0.35 raises completed fertility by 10% when a child requires half a room, and by 17% when it requires 1.6 rooms. The child's space must be paid in every period the child is at home, including after a bad earnings draw, when consumption is most valuable. The more space a child requires, the more these periods cost, and the more a household gains from lower risk.

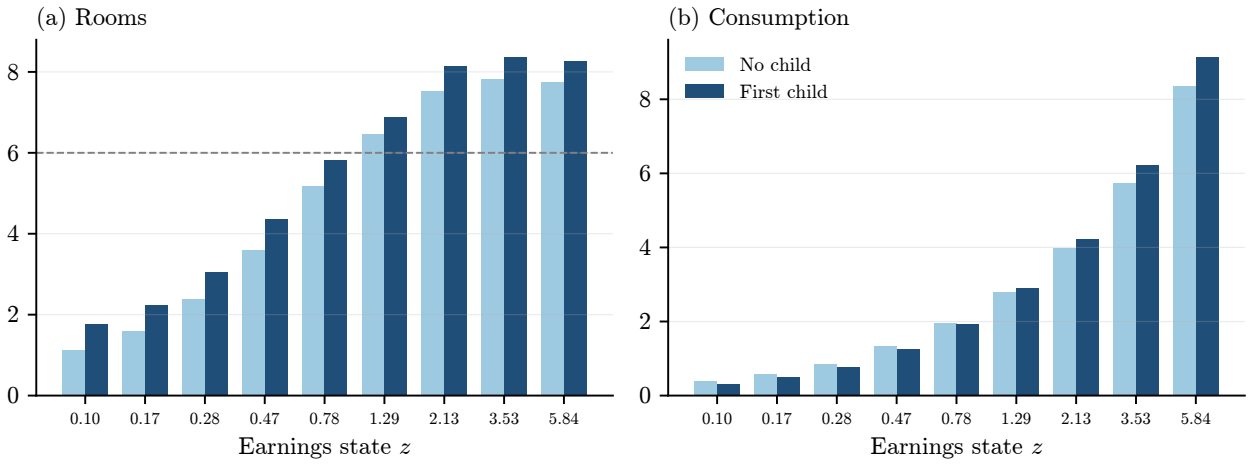


Figure 13: Choices with and without a first child, by earnings

Notes: Renters aged 26–29 without savings, 2007 stationary equilibrium, by earnings state z. Rooms and consumption are averages over tenure choices. The dashed line marks the rental cap $\bar{h}^R$.

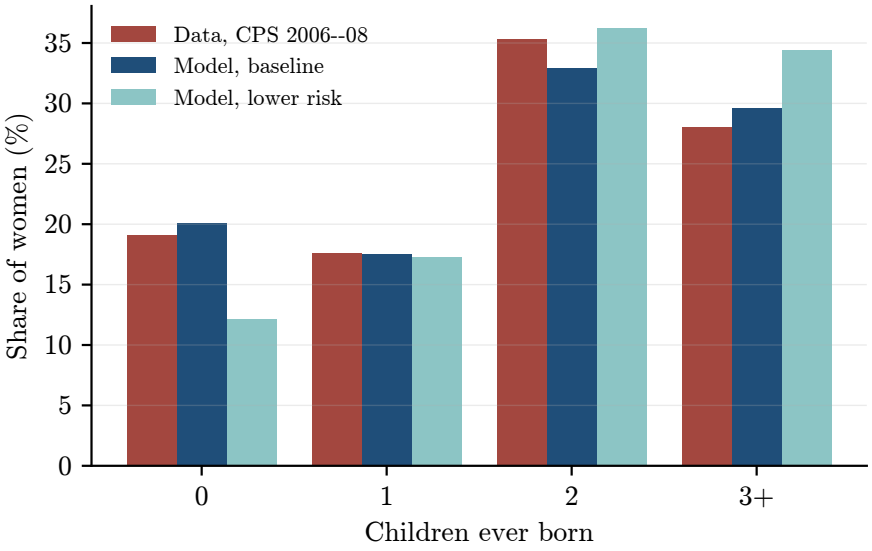


Figure 14: Earnings risk and children ever born

Notes: Shares with 0, 1, 2, and 3 or more children ever born. Data: CPS June Fertility Supplement, 2006 and 2008, women aged 40–44. Model: households at about age 42, 2007 stationary equilibrium, at the estimated earnings risk ($\sigma_z = 0.48$, baseline) and at lower risk ($\sigma_z = 0.35$, earnings states rescaled to keep mean earnings unchanged), same price.

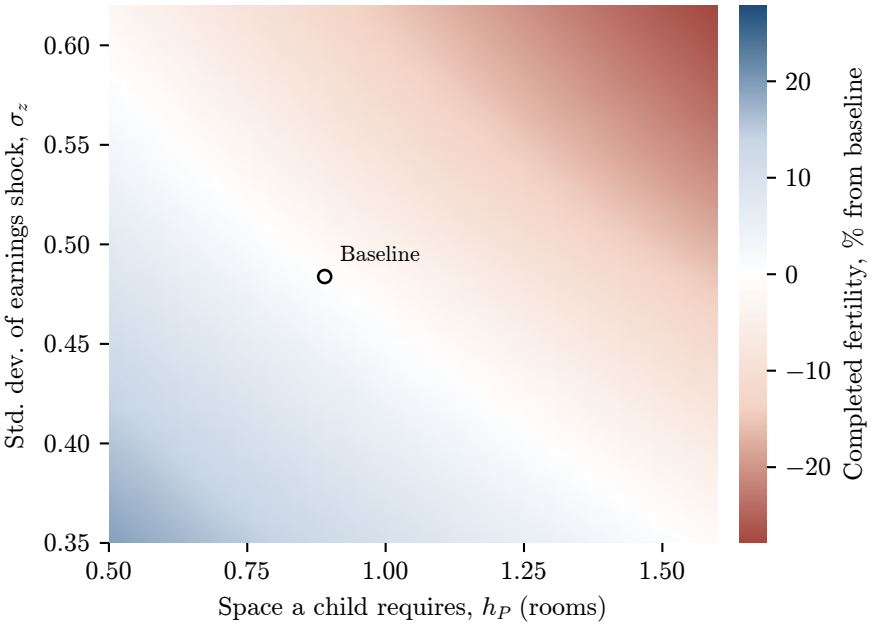


Figure 15: Earnings risk and the space a child requires

Notes: Completed fertility, % change from the baseline (marked), 2007 stationary equilibrium, same price, on a grid of 25 values of each parameter. Vertical axis: standard deviation of the four-year earnings shock, with earnings states rescaled to keep mean earnings unchanged. Horizontal axis: space a child requires, h_P, in rooms.

6 Counterfactuals

In this section, I use the calibrated model to study counterfactual changes to the 2023 economy and unpack the interaction between family space and fertility. First, I show that relaxing credit and expanding the housing supply raise births by the same amount on impact but have different effects over time, with housing supply having a stronger long-run effect. Second, I show that the main reason credit moves births earlier, but doesn't have as strong of a permanent effect, is because young households use it to buy the small homes they would otherwise rent. Intuitively, credit lowers the down payment, not the cost of the space a child needs. Third, I show that credit tied to children, which does lower that cost, raises the completed fertility of the 2023 cohort by 4.9%, and I compare these changes with a property tax.

6.1 Credit relaxation and housing supply

I compare two changes that intervene in the housing market. The first relaxes credit: the loan-to-value limit ϕ rises from 80% to 100%, so buyers no longer need a down payment. As Section 3.5 shows, credit raises the probability of trying for a child only if it is worth more to a household with the additional child than without it (3.13). Importantly, it can raise fertility when the down payment stands between a household and the home its family needs. The quantification is hence an important tool to assess whether there is a significant mass of agents that are giving up on having kids due to their inability of affording a 'family home' with many rooms. The second change expands the housing supply: the scale of the supply curve (3.9), $\bar{H}$, rises by 4.35%, so that at any price the stock is 4.35% larger. I choose this size so that births in 2023 rise by as much as under the credit policy. Credit lowers the down payment needed to buy a home. Supply lowers the cost of housing, both the price and rents. With the same effect on impact, the comparison shows which of the two costs matters for fertility.

Both experiments start from the 2023 state of the transition in Section 4.4. In 2023 the change is announced unexpectedly and understood to be permanent by agents. I compare each experiment with the path the economy follows from the same state without the change. The preference for children follows the same path with and without the change. Appendix H repeats the experiments with a more elastic supply and with a slowly adjusting stock.

I report births in the four-year periods starting in 2023 and 2063 and in the long run, when the economy reaches its new stationary equilibrium, and the completed fertility of the cohort aged 18 to 21 in 2023: the children ever born to these households over their lives. All are relative to the no-policy path. Table 4 summarizes the results; it also reports credit for parents, which Section 6.4 describes.

Table 4: Counterfactuals from 2023

	Births			Completed fertility	House price	
	2023	2063	Long run	2023 cohort	2023	Long run
Credit relaxation	+1.3	+0.6	+3.0	−1.1	−0.1	+3.2
Housing supply, +4.35%	+1.4	+3.0	+4.3	+2.0	−3.9	0.0
Credit for parents	+5.1	+16.6	+36.1	+4.9	+0.3	+33.5

Notes: % differences from the no-policy path from the same 2023 state. Births: births in the four-year period starting in the year shown; long run: the new stationary equilibrium, where births per household are at replacement, so births and the number of households rise by the same percentage. Completed fertility: children ever born to the cohort aged 18–21 in 2023. Without policy, period fertility is 1.63 in 2023 and 1.73 in 2063. Housing supply: the supply curve shifted out so that births in 2023 rise as much as with the credit relaxation.

6.2 Same impact, different long run

Figure 16 plots births and the number of households under the two changes. To understand the dynamics at play, it's helpful to look at the effects at different time horizons.

On impact, in 2023, the two changes look alike: both raise births by about 1.4%. Immediately after that, though, the path of the credit relaxation scenario displays the signs of a 'harvesting' effect: when the down payment disappears, many households act at once, and birth rates rise at every fertile age. But most of these households only bring forward children they would have had anyway.

In the medium run, over the four decades to 2063, the two paths separate. Under the credit policy, the impact gain is gone within four years, and in the following decade births fall below the no-policy path, by 0.6% at the bottom in 2035: the children born early are missing later. Births then recover, to about 1% above the no-policy path by 2055, as households that enter later gain more from credit while housing becomes cheaper. Under the supply policy, births keep rising, to 3.0% above the no-policy path by 2063. Two forces move births after impact: how many children each household has, and how many households are of childbearing age. A policy that raises births today raises the second about twenty years later, when the additional children reach adulthood, form households of their own and have children in turn: a demographic momentum (Keyfitz, 1971), or an 'echo' of the earlier births. Under the supply increase, birth rates stay higher for decades, since every cohort can buy cheaper housing, and from about 2040 the echo adds to them; by 2063 the two contribute about equally. Under the LTV policy, the births brought forward and the births missing later roughly cancel, so few additional households are born and the echo stays small. The cohort aged 18 to 21 in 2023 shows the difference (Table 5): in the supply increase scenario it has 2.0% more children over its life, and in the credit relaxation scenario 1.1% fewer.[25] Credit changes when households have

[25]The 2023 cohort ends with *fewer* children than without the policy because of the age composition of the population it retires into. After the fast decline, each cohort that reaches working age is smaller than the one before, so the population gradually ages. In 2071, the age mix is at its worst: the original larger cohorts are all retired, and the workforce consists of the younger, smaller ones, with a particularly unfavorable demographic balance. Retirees per worker rise from 0.35 in 2023 to 0.47 in 2071. After that, the population keeps shrinking but grows younger again: fertility recovers toward replacement (from 1.63 in 2023 to 1.86 in 2087), each new cohort is only a little smaller than the last, and retirees per worker return to 0.35 in the long run. With a fixed payroll tax, the pay-as-you-go pension

children more than how many they have.[26]

After 2055, births with the credit relaxation fall back a second time, to 0.25% above the no-policy path in 2071, before the echo lifts them toward the long run. This second decline reflects how the effect of credit depends on the price of housing. When homes are very expensive, credit mainly lets young couples buy earlier and have their first child earlier; when homes are particularly cheap, it has a stronger effect in pushing towards larger houses; in the intermediate price range, which is particularly salient for cohorts in their thirties after 2055, neither channel is strong, and credit mostly turns childless renters into owners of mid-sized homes who end up with about as many children (Appendix I.1).

In the long run, the supply expansion raises the number of households, and births with it, by 4.3% at an unchanged price. The credit relaxation raises them by 3.1%, with a price 3.2% higher.[27]

The two paths help us understand an important concern when thinking about fertility policy. Births brought forward are missing later, so a policy that mostly moves timing adds few births over time. A policy that makes children cheaper at a given price adds births in every cohort, and the echo of those births compounds them: by 2063, births are 3.0% higher with the supply expansion and 0.6% higher with the credit relaxation.

Table 5: The cohort aged 18–21 in 2023

	Completed fertility	Share with a first birth	Mean age at first birth
No policy	1.640	0.673	27.15
Credit relaxation	1.622 (−1.1%)	0.665 (−1.3%)	27.07
Housing supply	1.673 (+2.0%)	0.684 (+1.6%)	27.05
Credit for parents	1.720 (+4.9%)	0.701 (+4.1%)	26.88

Notes: Children ever born and the share with a first birth, per household of the cohort, summed over ages 18 to 45. Mean age at first birth: first-birth rates weighted by the midpoints of the four-year age cells.

follows this age composition: it falls by a quarter by 2071 and then recovers. The 2023 cohort retires around 2071, at the lowest point, and with the credit relaxation its households enter that period having bought earlier and with more debt. Anticipating this, they have fewer children. With the pension held at its 2023 level, the same relaxation raises this cohort's completed fertility by 0.6%. Later cohorts gain because house prices keep falling while they raise children: with the price held at its 2023 level, every cohort has fewer children. See Appendix I.1 for more detail.

[26] Demographers call these the *tempo* and the *quantum* of fertility. When households bring births forward, period fertility can rise immediately and decline later, even if the number of children of each cohort remains unchanged. Bongaarts and Feeney (1998) accordingly split the period total fertility rate into a quantum component, "the TFR that would have been observed in the absence of changes in the timing of childbearing," and a tempo component, "the distortion that occurs due to timing changes." Kearney and Levine (2026) use the same distinction to argue that the recent fall in fertility in high-income countries has to be read from the completed fertility of cohorts, which is the measure I use here.

[27] In the long run births per household are at replacement, so births and households move together. The long-run effect of credit comes from a small rise in fertility at a given price, which is larger at the low long-run price, 45% below its 2023 level. Letting renters rent homes as large as 10 rooms removes about half of it.

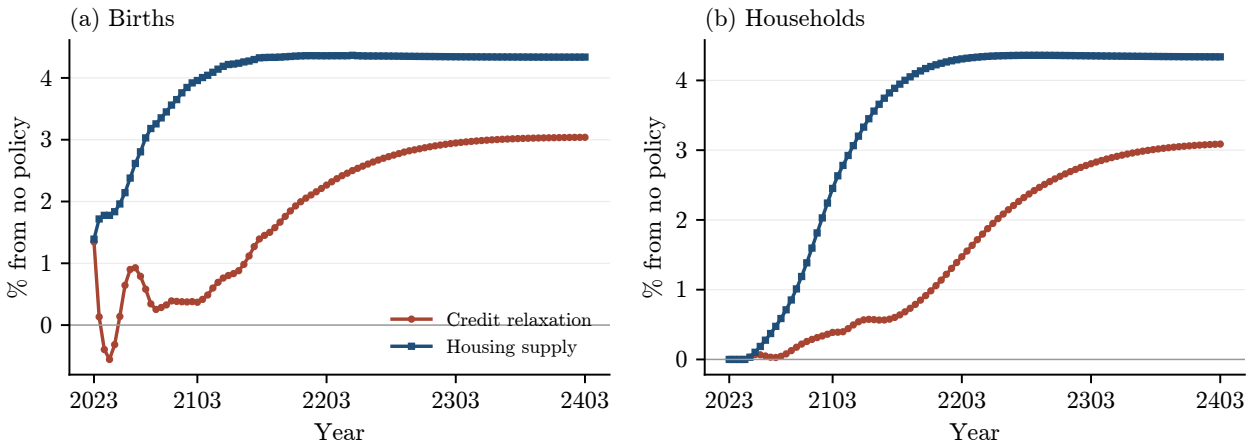


Figure 16: Credit relaxation and housing supply

Notes: % differences from the no-policy path from the same 2023 state, births in each four-year period (panel a) and the number of households (panel b), to the new stationary equilibrium. Credit relaxation: LTV 80 to 100 for all households. Housing supply: the supply curve shifted out by 4.35%.

6.3 Unpacking the credit policy

Why does credit move births earlier rather than raise their number? Intuitively, this policy lowers the equity needed to buy a home, $(1-\phi)P_t h^O$, but not the per-period cost of the space a child needs. Rent equals the user cost of owning (3.10), so a family can rent that space at about the per-period cost of owning it. By (3.13), credit raises the probability of trying for a child only if it is worth more to a household with the additional child than without it. To understand this better, I look in more detail at the response to the LTV relaxation policy.

Figure 17 shows where young households live in 2023. The credit relaxation turns a fifth of households aged 18 to 29 from renters into owners, and almost all of them buy small dwellings: the share owning a home of 2 rooms rises from 8% to 24%, and of 4 rooms from 13% to 17%, while the shares owning 6 rooms or more barely move. The supply expansion leaves the housing of this age group almost unchanged.

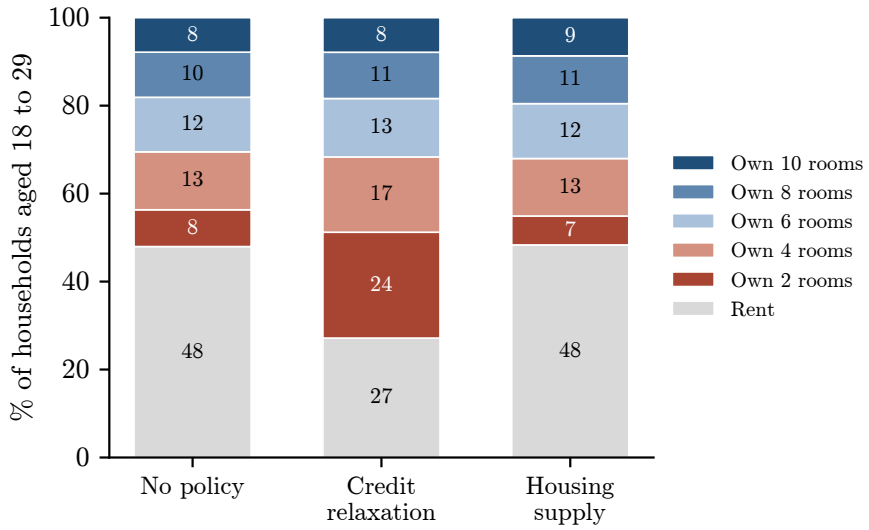


Figure 17: What young households live in, 2023

Notes: Households aged 18–29 in 2023, after the tenure choice, renting or owning a home of 2 to 10 rooms, %. Each experiment starts from the same 2023 state as the no-policy path.

Figure 18 follows the cohort aged 18 to 21 in 2031 through its life. With the credit relaxation, the share owning a home of 2 or 4 rooms roughly doubles in its late twenties and stays well above the no-policy path into its forties (panel a), while the share owning a home of 6 rooms or more is about the same (panel b). Births come slightly earlier, and the cohort ends with slightly fewer children, 0.4% (panel c). The supply expansion instead raises the share in larger homes and births at every age, and the cohort has 1.8% more children. These results illustrate that the mechanism proposed in Section 3.5 does not seem to hold. Households do not use the newly found credit abundance to purchase big houses and have kids, but rather switch from rental to ownership of similar sized dwelling and, if anything, this discourages births,[28] as it makes it more likely these families do not want to expand over the limits of their bought house.[29]

[28]The PSID offers a descriptive check of this: among childless first-time buyers aged 22 to 33, those who bought a home of four rooms or fewer were 14 percentage points less likely to have a first birth within six years (standard error 6.7; Appendix I.3).

[29]This is in line with what Kaplan et al. (2020) find for the credit relaxation of the US housing boom. In their model, renters who become owners when credit is relaxed "buy houses similar in size to the ones they were previously renting," so that "the age profile of homeownership shifts to the left—households who would have become owners in the future buy houses today . . . —but the life-cycle profile of consumption of housing services remains virtually unchanged."

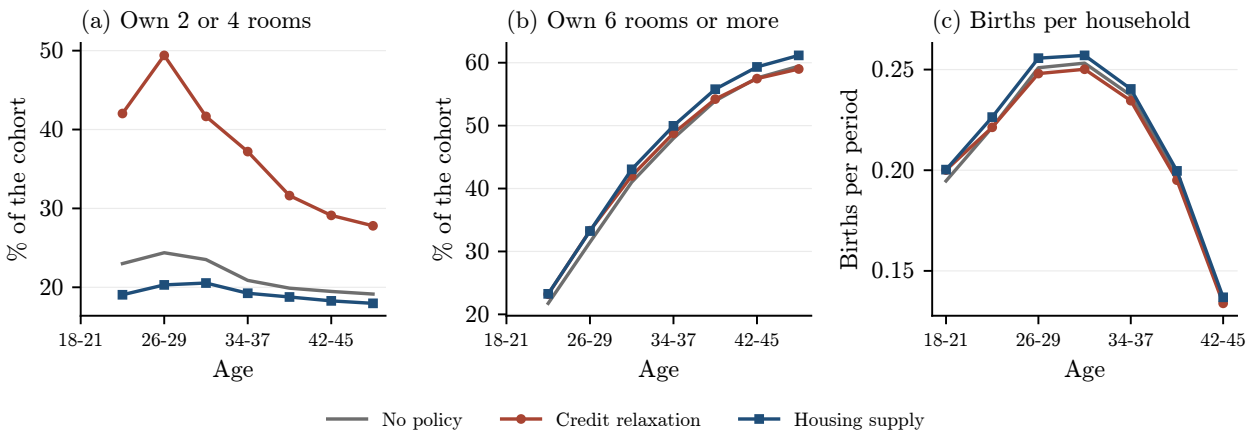


Figure 18: The cohort aged 18–21 in 2031: housing and births by age

Notes: Share of the cohort owning a home of 2 or 4 rooms (panel a) and of 6 rooms or more (panel b) at the start of each four-year age group, and births per household in each age group (panel c). Each experiment starts from the same 2023 state as the no-policy path.

Perhaps surprisingly, the modest effect of the credit relaxation does not seem to be due to a general-equilibrium feedback through higher house prices. When I hold prices, rents and transfers at their no-policy paths (Appendix I.1), the credit relaxation raises births in 2023 by about as much as in equilibrium, 1.4%, and the cohort aged 18 to 21 in 2023 still ends with fewer children.

6.4 Credit for parents

The previous sections seem to suggest that a housing policy aimed at increasing fertility should be one that shifts the relative value of parenthood. To see this, I study a very similar credit relaxation, but targeted to parents. I raise the loan-to-value ratio to 1 only for households with a child at home. While giving the same monetary benefit, the policy has the important difference of explicitly encouraging births. As a result, births in 2023 rise by 5.1%, several times as much as relaxing credit for all households, at nearly the same price, 0.3% higher. The completed fertility of the 2023 cohort rises by 4.9%, and the share with a first birth by 4.1% (Table 5). Births keep rising, by 17% by 2063 (Figure 19). As the larger cohorts demand more space, the price rises, by 3.5% by 2063 and 33% in the long run, when there are 36% more households. The significantly different results are due to the new value of having a child. A birth now unlocks ownership without a down payment, so in 2023 childless renters aged 22 to 29 try for a first child 13 to 16% more often. Once they have children, households use the credit: renter-parents aged 22 to 25 buy a home with probability 0.63 instead of 0.57.

6.5 Increasing property taxes

Coven et al. (2026) show that a higher property tax is capitalized into a lower house price, which lowers the down payment and makes it easier for young households to buy. This makes an increase

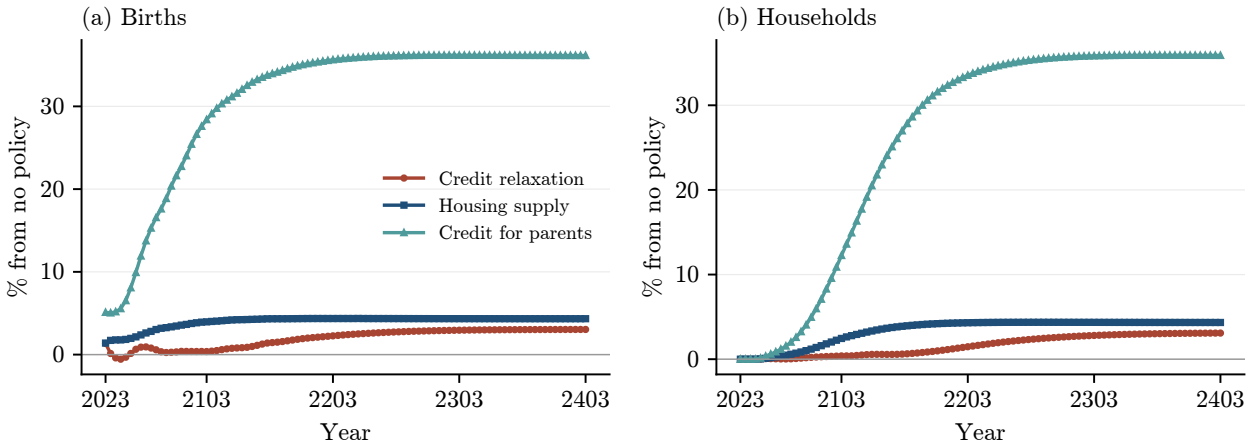


Figure 19: Credit relaxation, housing supply and credit for parents

Notes: % differences from the no-policy path from the same 2023 state, births in each four-year period (panel a) and the number of households (panel b), to the new stationary equilibrium. Credit for parents: LTV 100 for households with a child at home.

in property taxes a natural candidate for a pro-natality policy in our context. To see whether this channel matters for fertility, I raise the annual property tax from 1.06% to 2% of value from 2023 on, and return the revenue to households as an equal transfer every period.

The channel is there: the tax lowers the price by 7.0% in 2023, and ownership at ages 18 to 29 rises by 2.9 percentage points. But a lower price does not, in this case, mean really cheaper housing. The price falls because whoever holds a home now pays more tax every year, and rental companies pass the tax on to renters: per four-year period, the cost of holding a room rises from 18.0% to 22.1% of its price in 2023, so even at the lower price a room costs more per year, and rents rise by 14.5% (Appendix I.4). Buying gets easier, but the extra room for kids is actually more expensive. As with the credit relaxation, easier access to ownership is not what keeps families out of the space they need (Section 6.3), so births barely move on impact (+0.1%). Over the following decades births do rise, and by more than with the credit relaxation (2.3% against 0.6% in 2063, Figure 20), but the gain remains modest in the long run, 2.2%. The rise seems to be mostly attributable to the rebate. Because it is the same for every household, it moves resources from households in large homes, mostly high earners, to those in small homes, mostly low and middle earners, and the additional births come from the latter.[30]

The comparison with Coven et al. (2026) is not one to one, but the model confirms the channel they emphasize can have a modest effect on fertility: the tax lowers prices, more young households own, and fertility does go up.[31] However, the model suggests that a higher property tax raises fertility

[30]When the additional revenue is instead returned in proportion to the rooms held by households of each age and earnings level, so that it no longer redistributes across earnings, the same tax lowers births by 3.3% in the final steady state even though the house price falls more. Without the redistribution, the higher cost dominates.

[31]In their model the tax falls only on owner-occupied housing: rentals are a separate market with their own supply, so the tax lowers the price young buyers pay without raising rents. Here instead it applies to rentals too, through the user cost being passed through by rental companies. Their setting is also spatially focused on Texas vs California rather than the whole United States. Older owners face a payment-to-income limit, capital gains taxes and the step-up

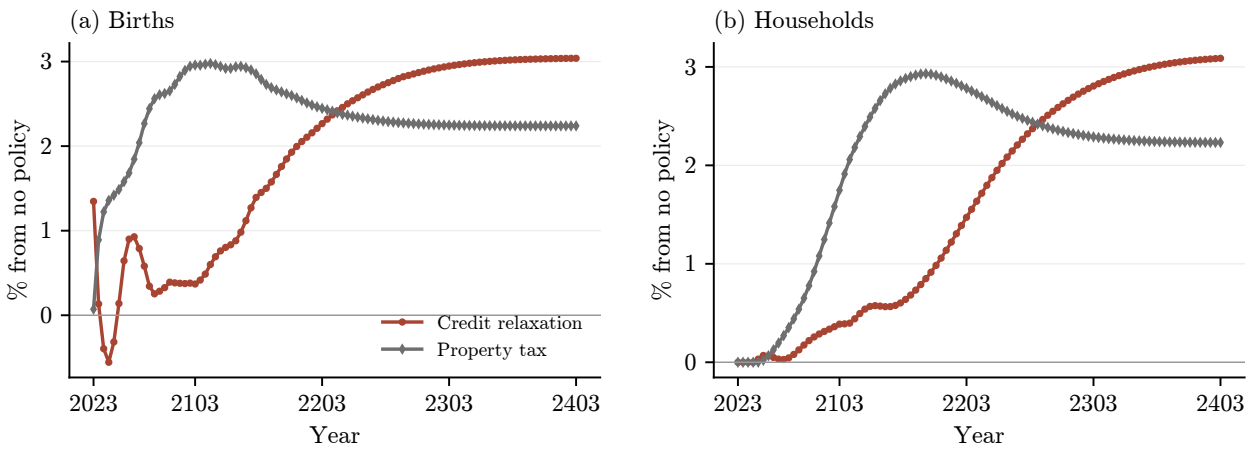


Figure 20: Increasing property taxes

Notes: % differences from the no-policy path from the same 2023 state, births in each four-year period (panel a) and the number of households (panel b), to the new stationary equilibrium. Property tax: from 1.06% to 2% of value a year from 2023 on, revenue returned equally to all households every period. Credit relaxation: LTV 80 to 100 for all households.

only if its revenue is returned in a way that favors the households about to have children, since the tax itself makes the space a child needs more expensive every year.

7 Conclusion

The decline in fertility has drawn widespread attention from the media, policymakers and academics, and many governments look to housing policy to raise births. Understanding the effects of these policies, and how childbearing and homeownership interact, is increasingly important. I study both in a life-cycle model in which households choose whether to have children, whether to rent or own, and how much to save, subject to earnings risk and a down payment constraint. In the model, children raise a household's needs for both consumption and space. This cost weighs most on young households with few resources, and it interacts with earnings risk.

The model shows the need to distinguish short-run effects, which may only reflect changes in timing, from longer-run effects. Specifically, letting all buyers borrow the full price of a home raises births by 1.3% in 2023, but young households mostly use the credit to own earlier, in relatively small homes, and the cohort aged 18 to 21 in 2023 ends up with no more children. Expanding the housing supply by enough to have the same effect in 2023 lowers housing costs by more, and births are 3.0% higher by 2063. Credit restricted to parents raises the completed fertility of the same cohort by 4.9%.

These results matter for the design of housing policy as fertility declines. As the population starts to shrink, falling house prices are a natural stimulus to fertility. In demography, however, effects

of basis at death, which keep them in large homes, and residents value the rebated revenue at less than its cost.

compound, so a policy can matter a great deal even if it only speeds up the recovery. The idea that house prices alone could reverse the decline in fertility is stark, but most of the results of this paper would also apply to a slowdown of the decline, even one that never brings fertility back to replacement.

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